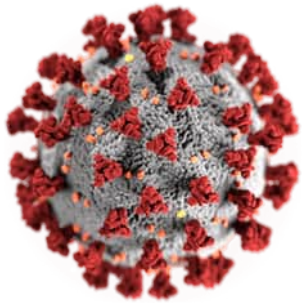
EECS 498-007 / 598-005
Deep Learning for Computer Vision

Lecture 1: Introduction

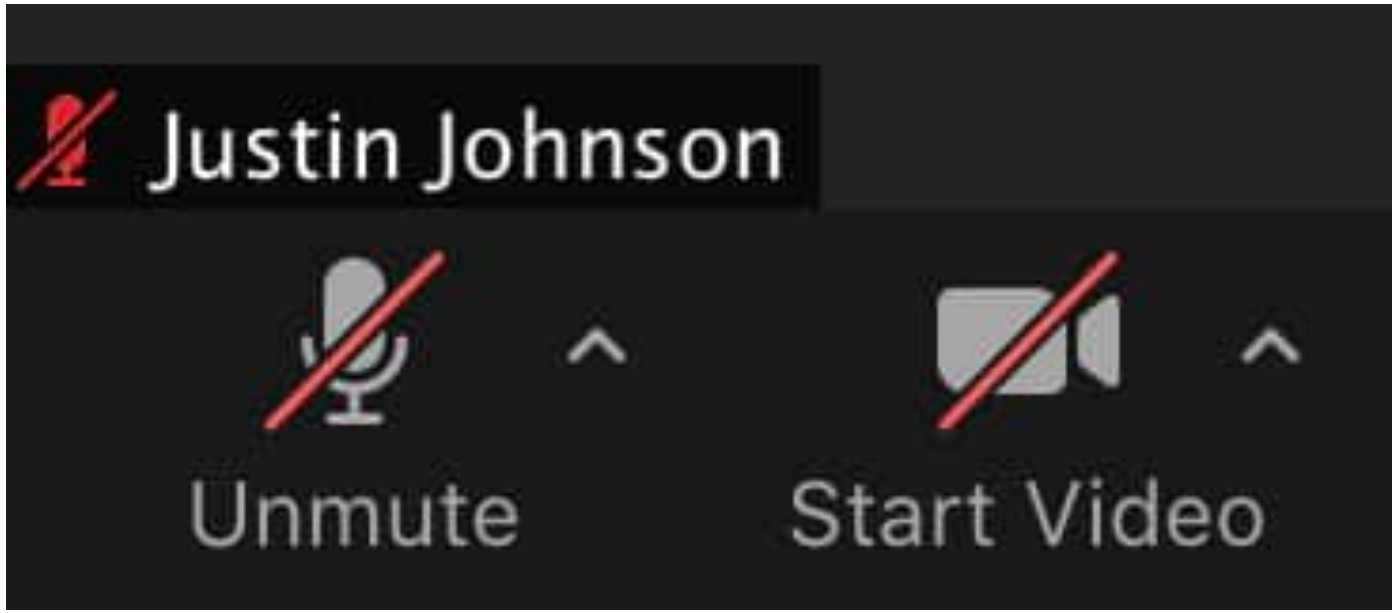
COVID-19 Edition



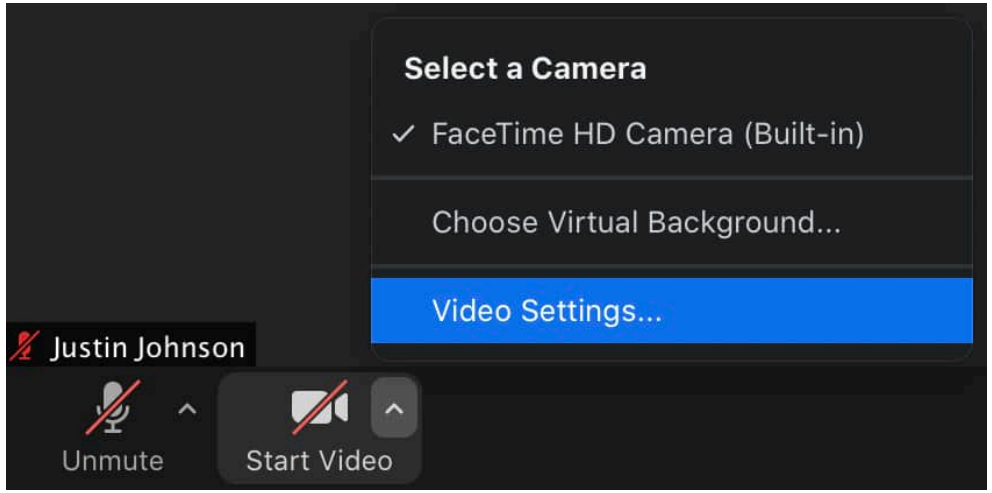
EECS 498-007 / 598-005
Deep Learning for Computer Vision

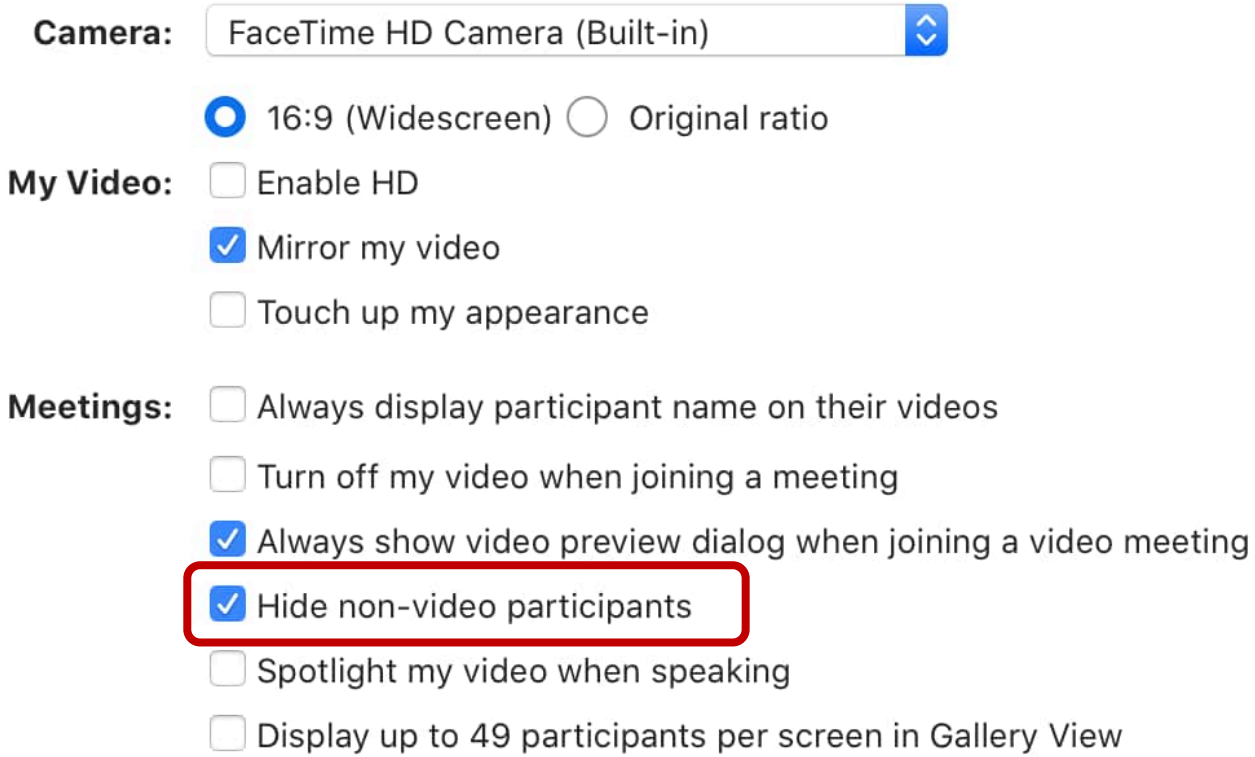
Lecture 1: Introduction

Zoom Logistics: Muted, Camera Off



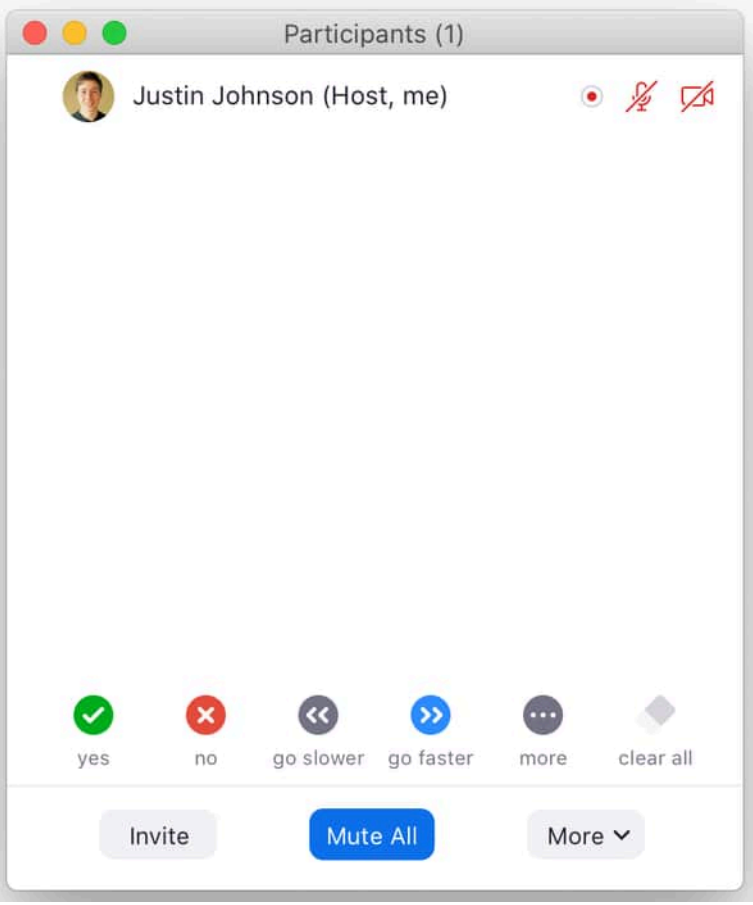
Zoom Logistics: Hide non-video participants





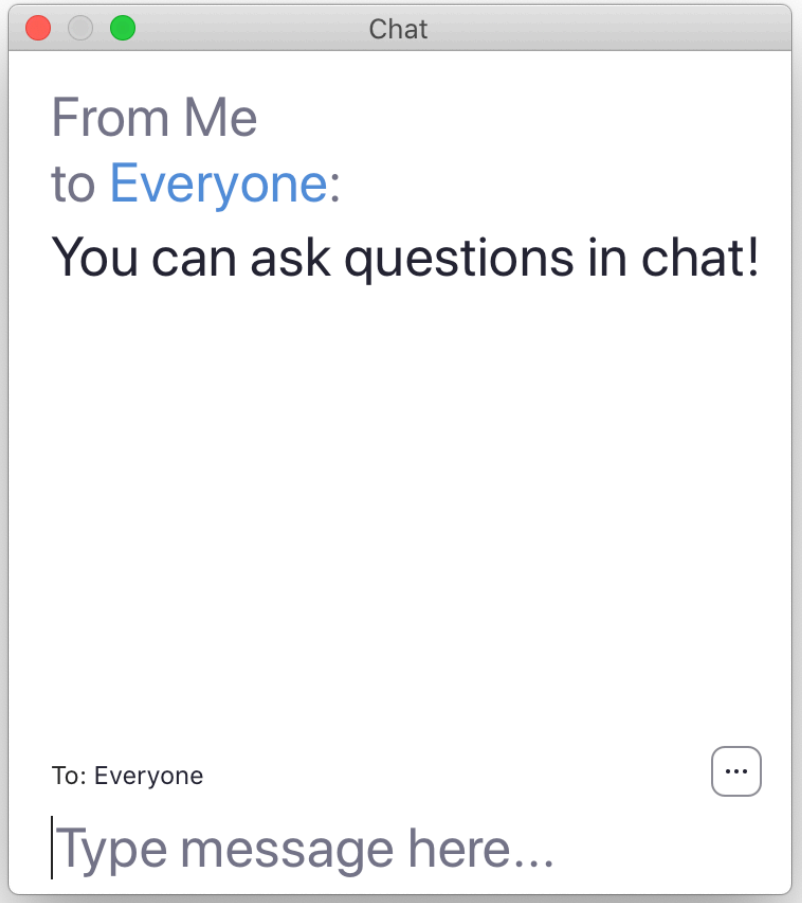
Zoom Logistics: Quick Poll

- You can use responses in ”participants” panel for quick polls

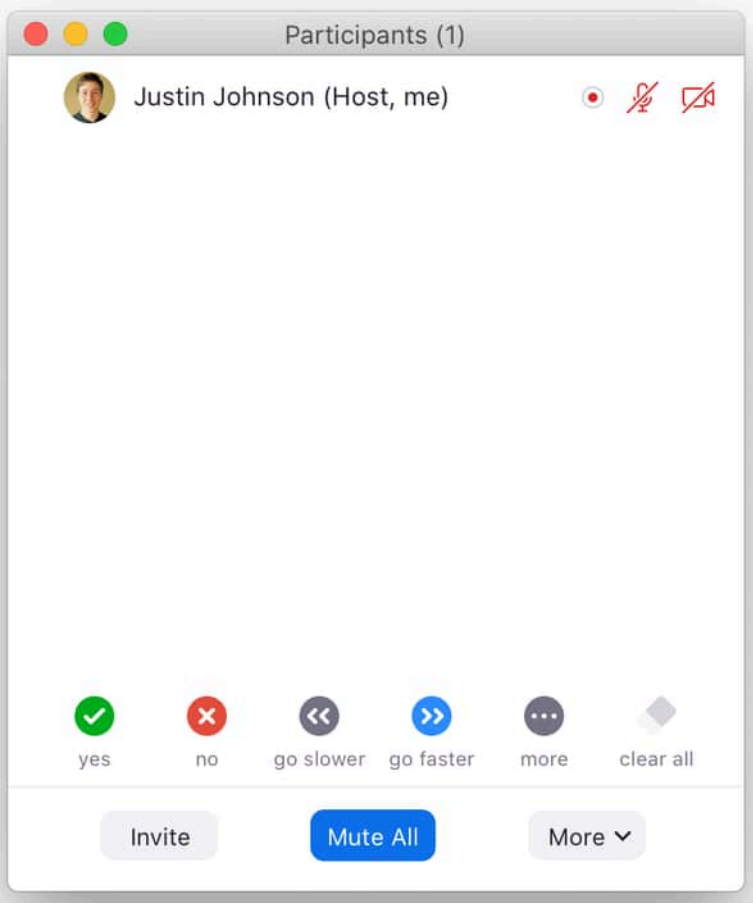


Zoom Logistics: Asking Questions

Option 1: Ask questions in chat



Option 2: “Raise Hand” in participants



Deep Learning for Computer Vision

Deep Learning for Computer Vision

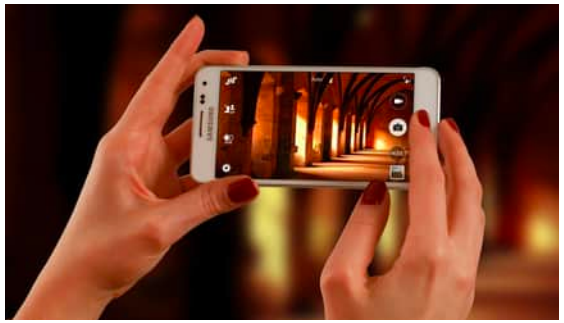
Building artificial systems
that process, perceive, and
reason about visual data

Computer Vision is everywhere!






Left to right:
Image by Roger H Goun is licensed under CC BY 2.0
Image is CC0 1.0 public domain
Image is CC0 1.0 public domain
Image is CC0 1.0 public domain






Left to right:
Image is free to use
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Image by NASA is licensed under CC BY 2.0
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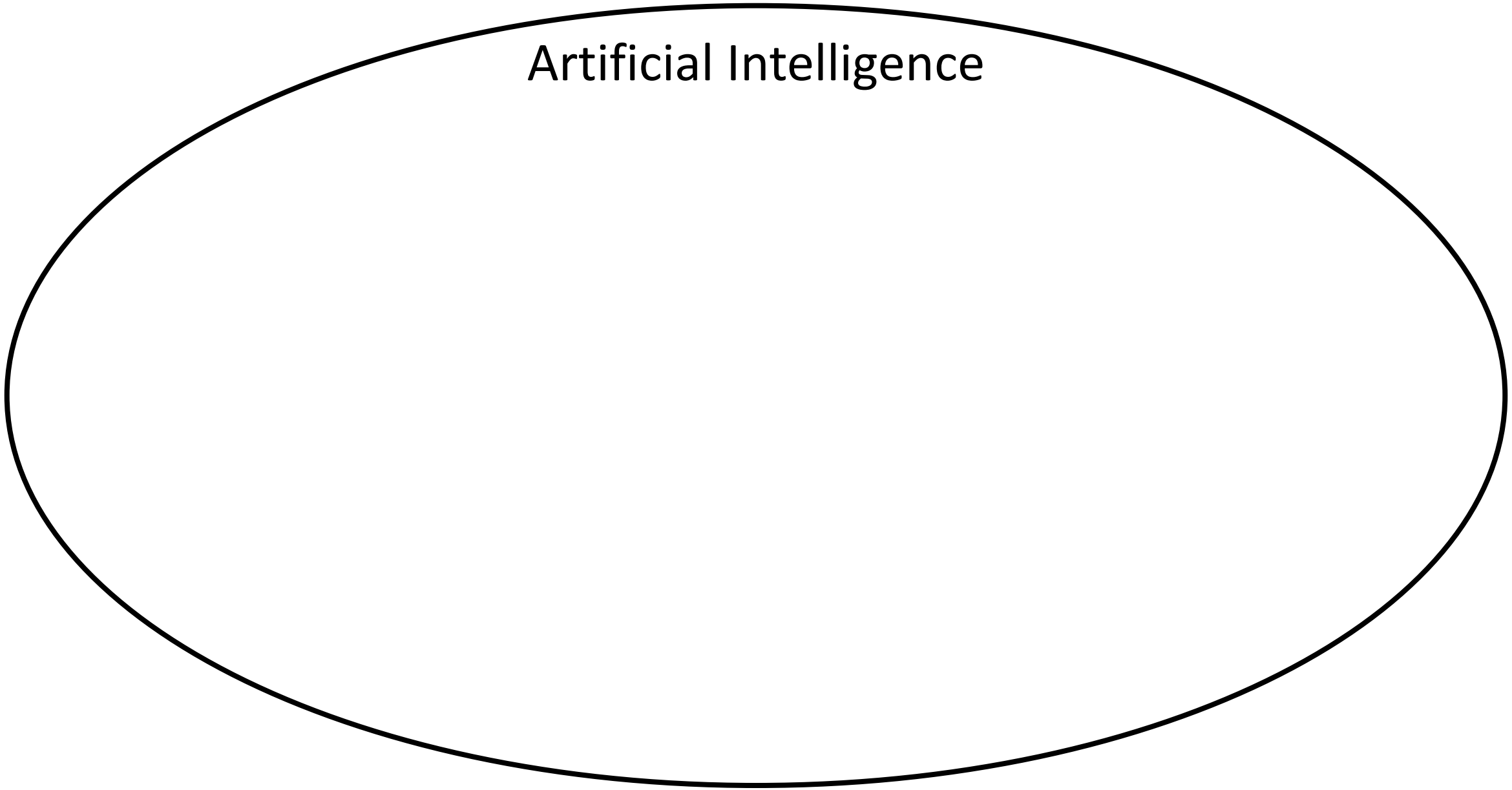

Bottom row, left to right
Image is CC0 1.0 public domain
Image by Derek Keats is licensed under CC BY 2.0; changes made
Image is public domain
Image is licensed under CC-BY 2.0; changes made

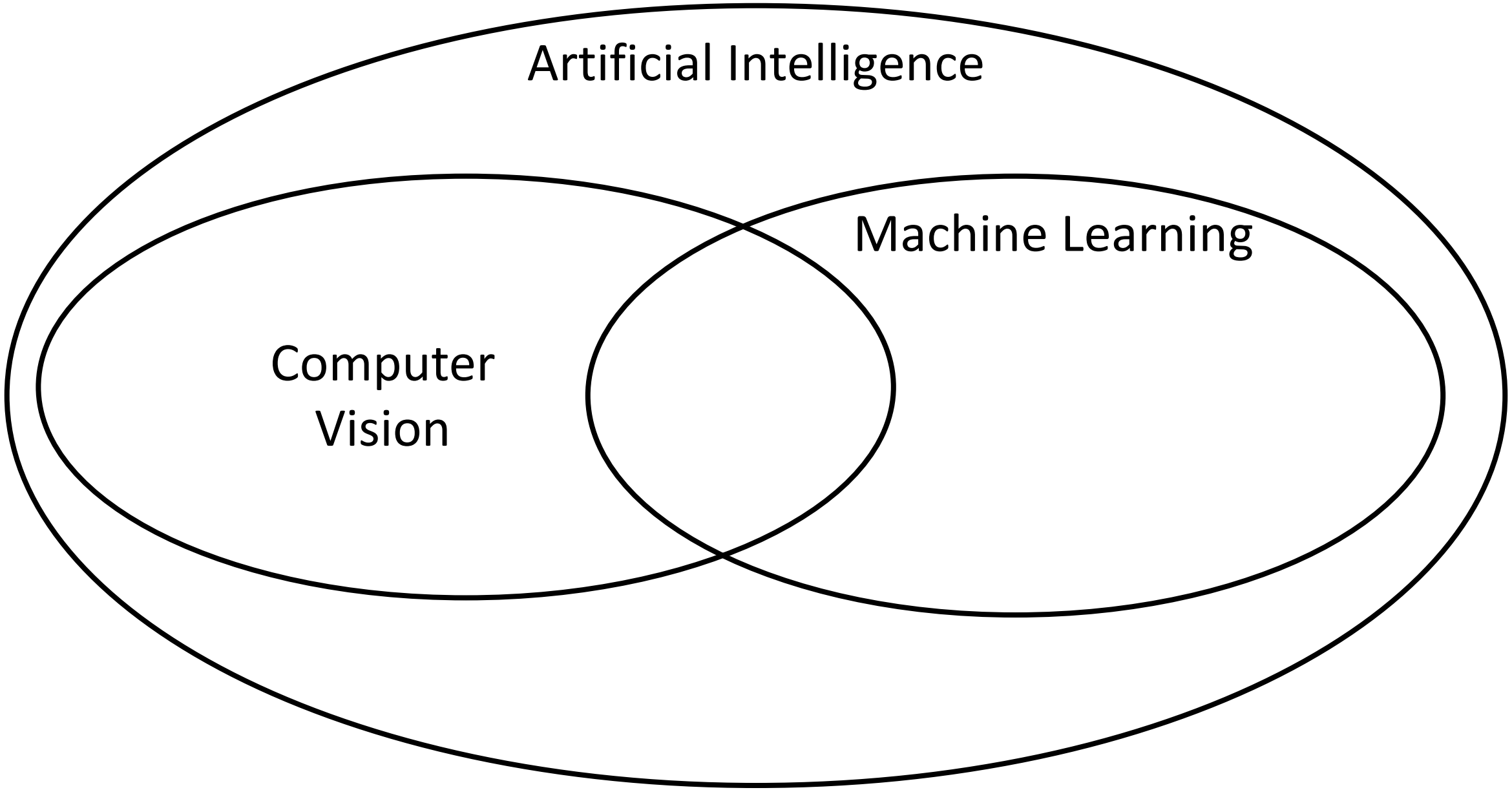
Deep Learning for Computer Vision

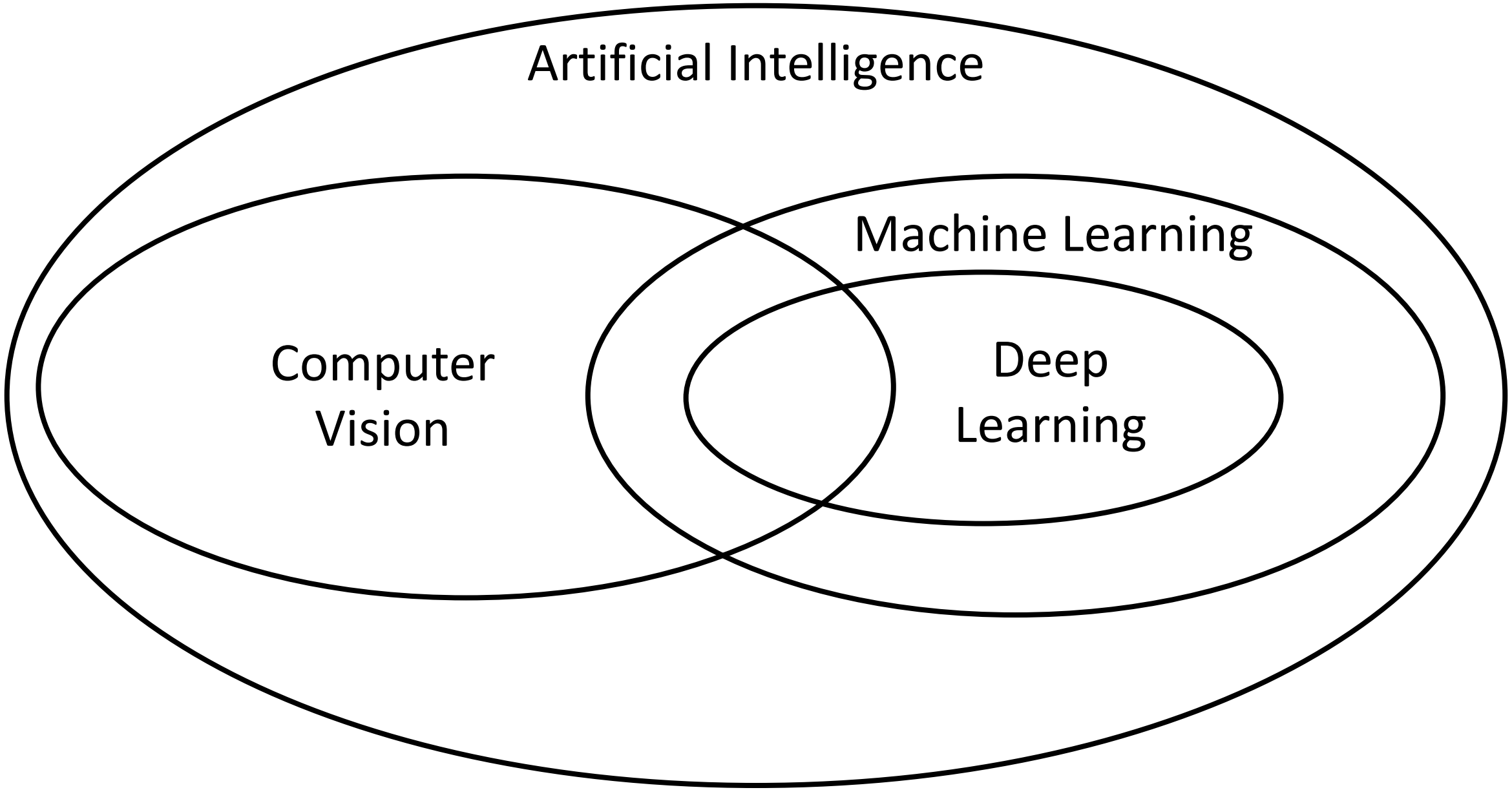
Building artificial systems that
learn from data and experience

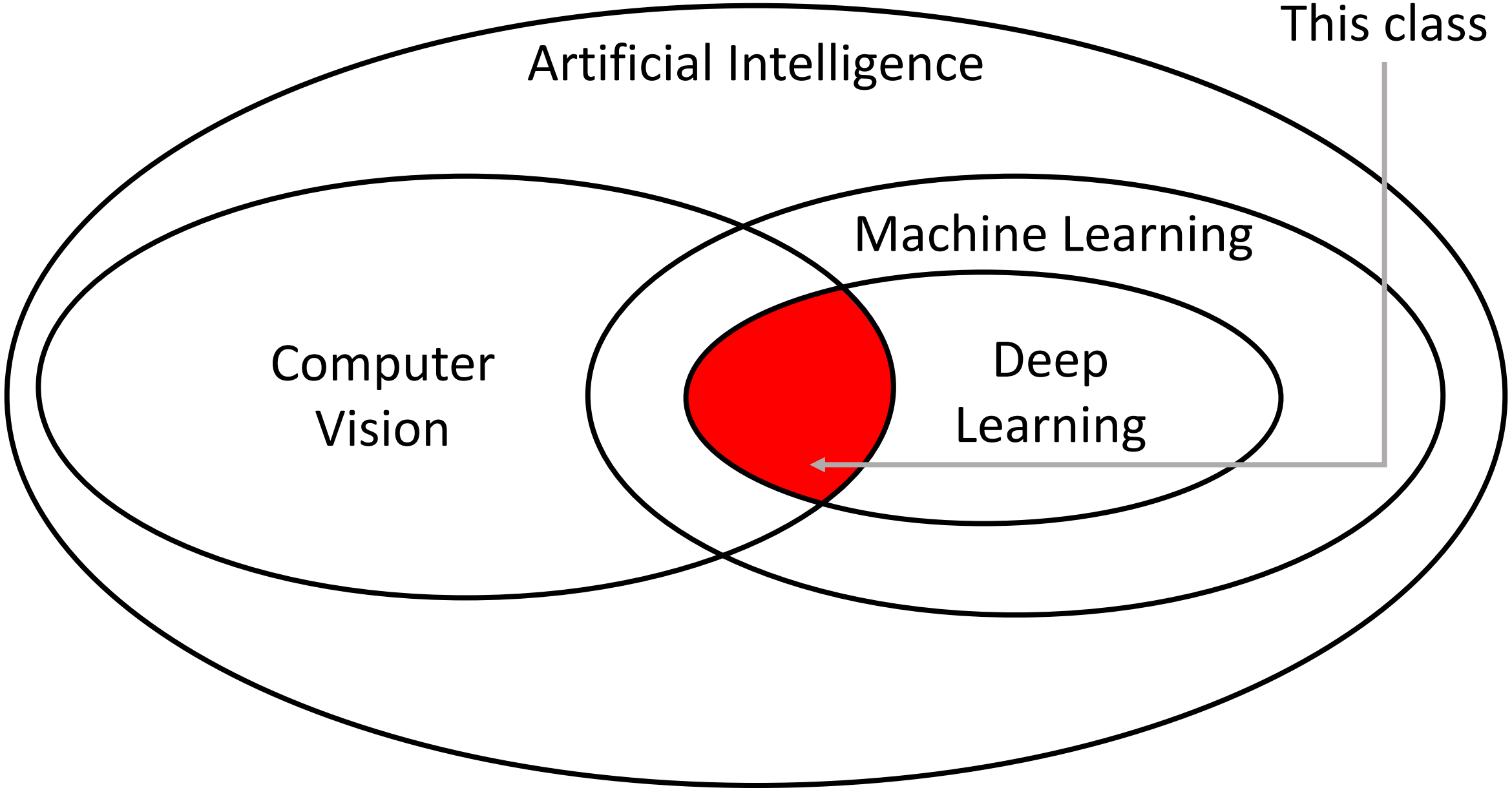
Deep Learning for Computer Vision

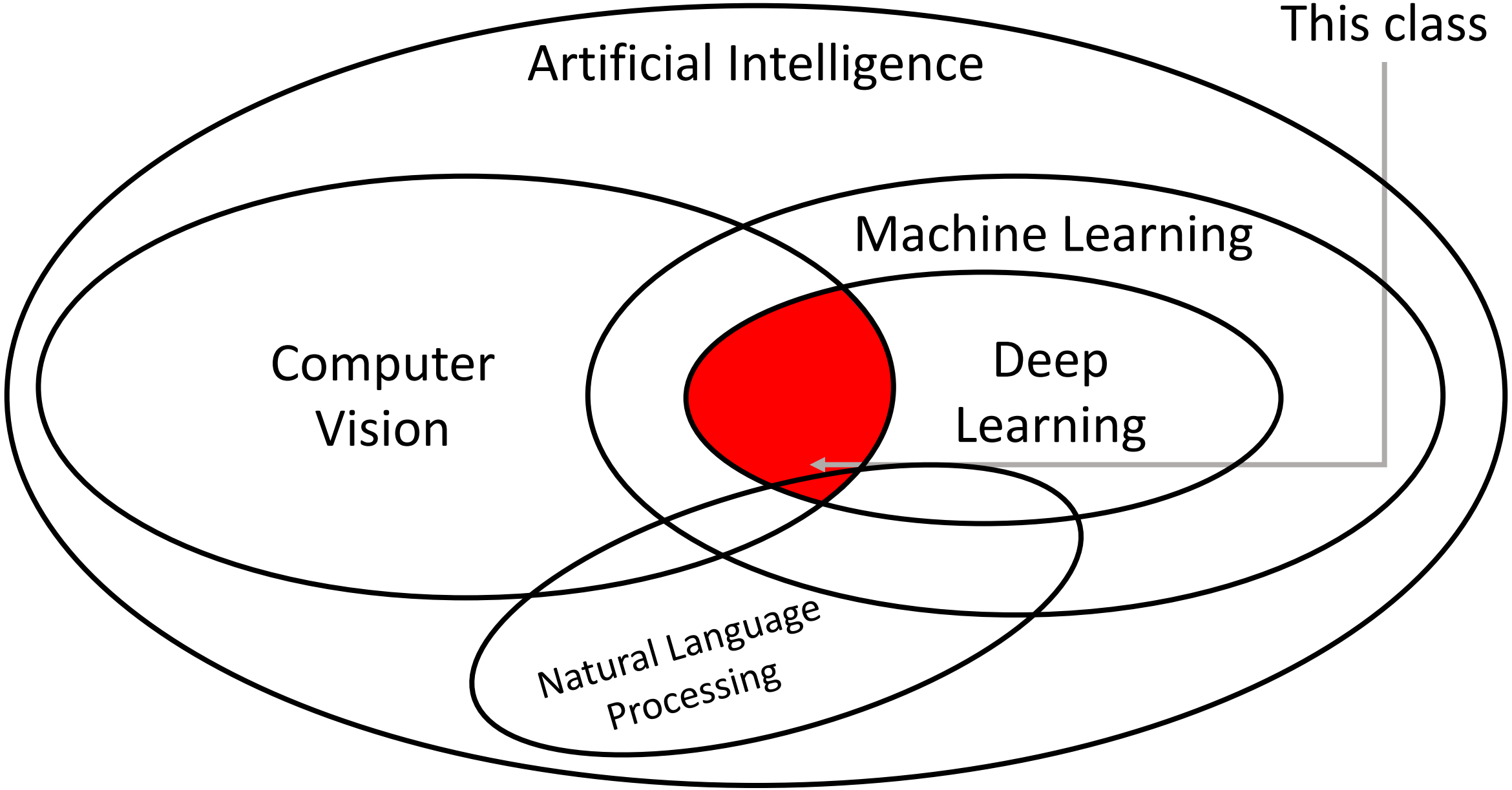
Hierarchical learning algorithms with many “layers”, (very) loosely inspired by the brain

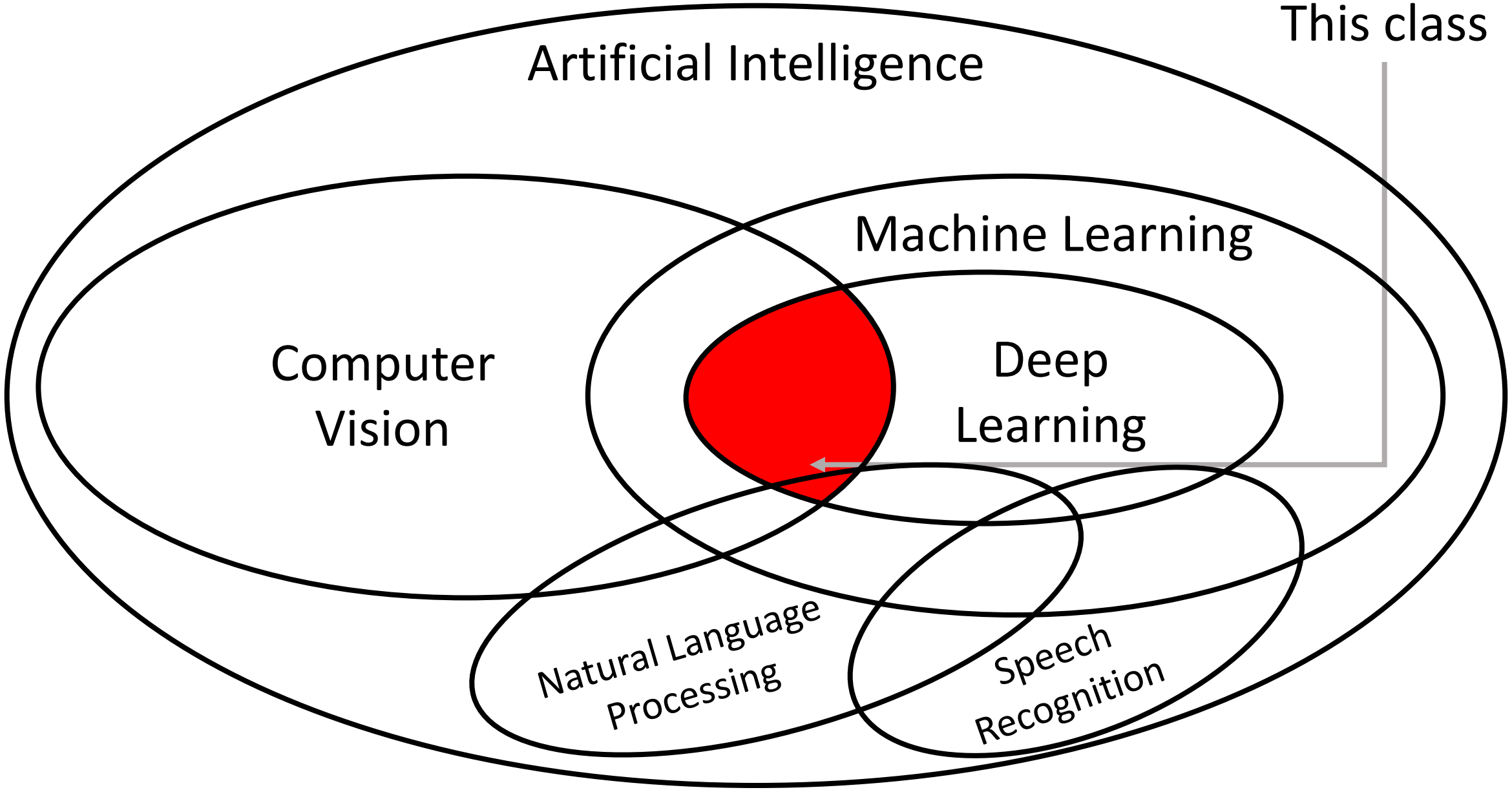

Artificial Intelligence

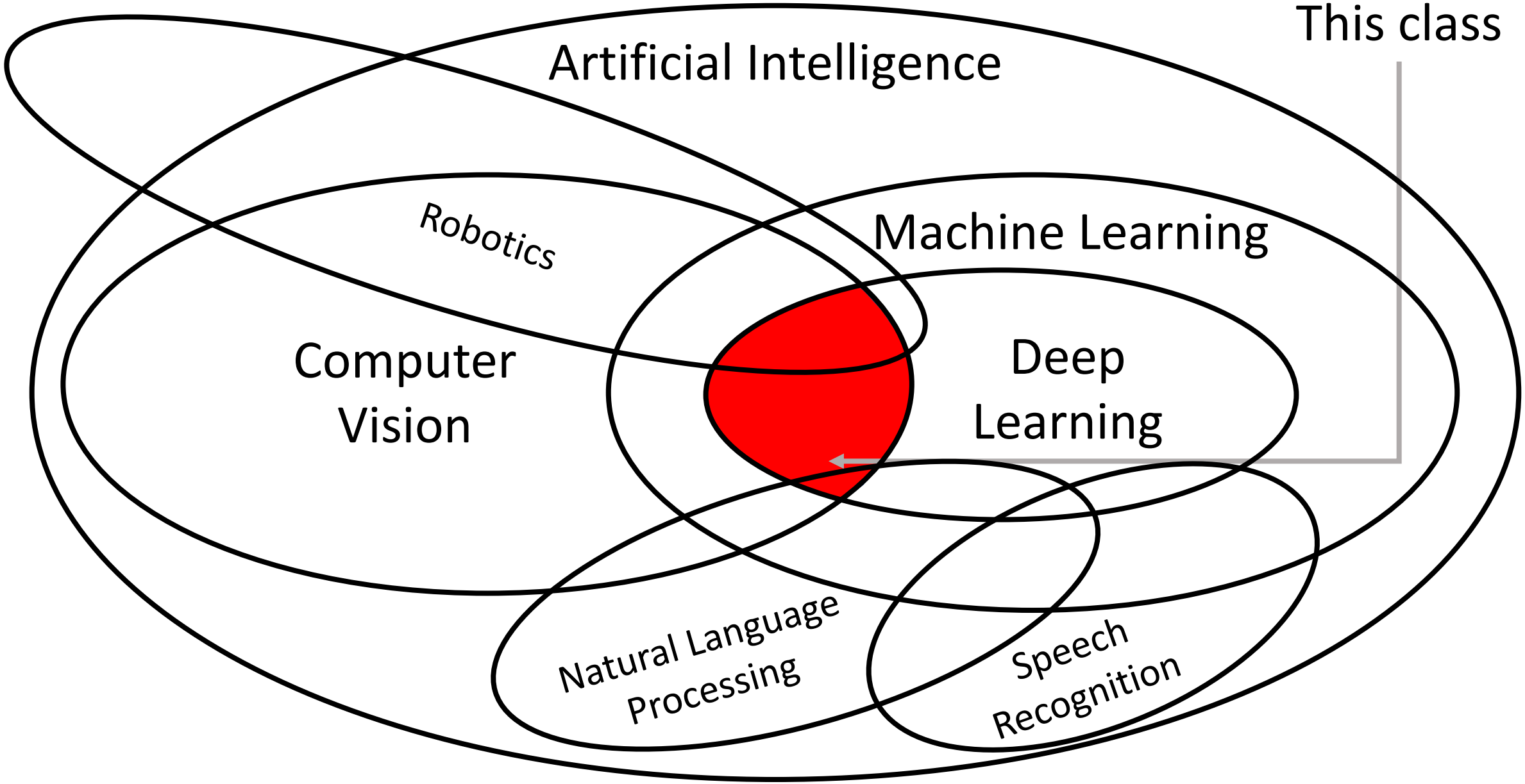

Artificial Intelligence
Machine Learning
Computer
Vision


Artificial Intelligence
Machine Learning
Computer Vision
Deep Learning


This class
Artificial Intelligence
Machine Learning
Computer
Vision
Deep
Learning


This class
Artificial Intelligence
Machine Learning
Computer Vision
Deep Learning
Natural Language Processing


This class
Artificial Intelligence
Machine Learning
Computer Vision
Deep Learning
Natural Language Processing
Speech Recognition


This class
Artificial Intelligence
Robotics
Machine Learning
Computer Vision
Deep Learning
Natural Language Processing
Speech Recognition

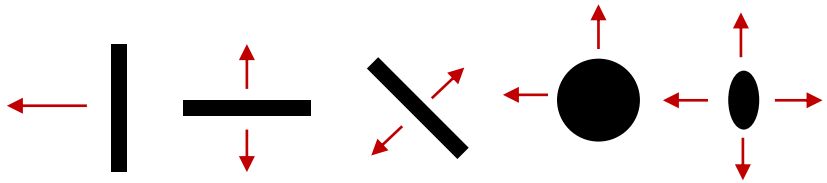
Today's Agenda

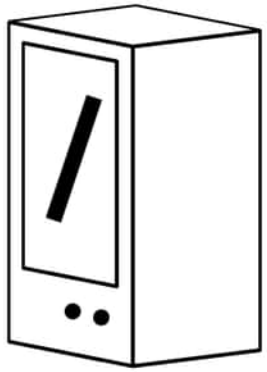
- A brief history of computer vision and deep learning
- Course overview and logistics

Hubel and Wiesel, 1959

Measure brain activity



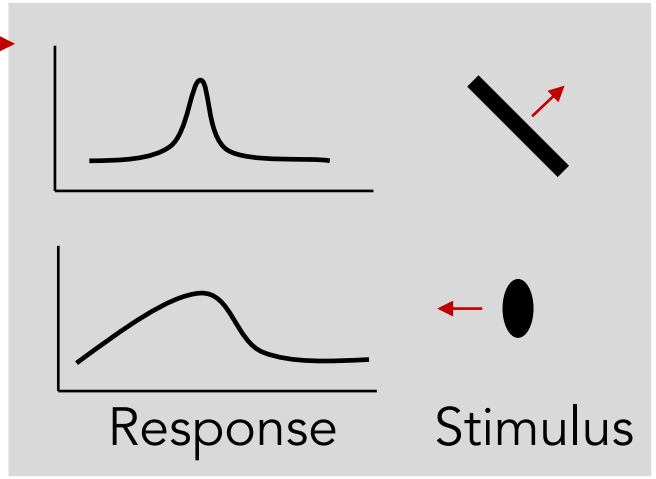


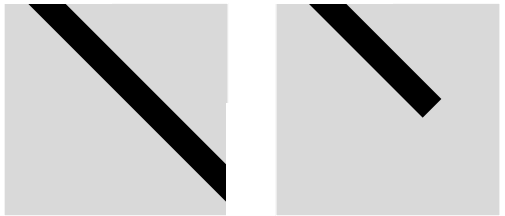


Simple cells:
Response to light orientation

Complex cells:
Response to light orientation and movement

Hypercomplex cells:
response to movement with an end point



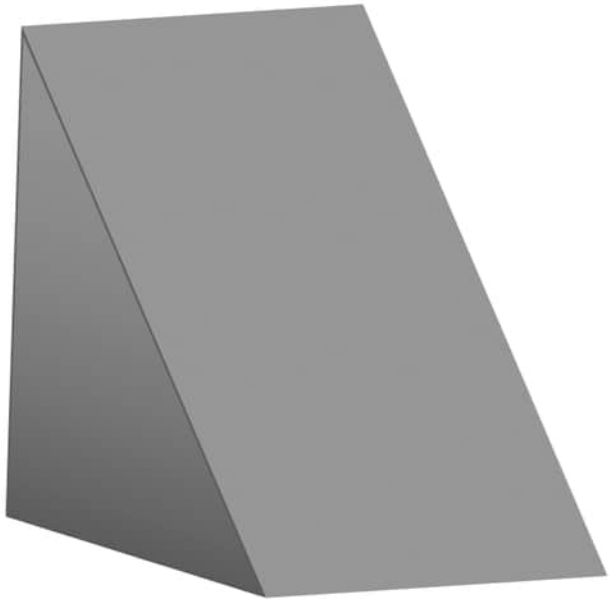


No response

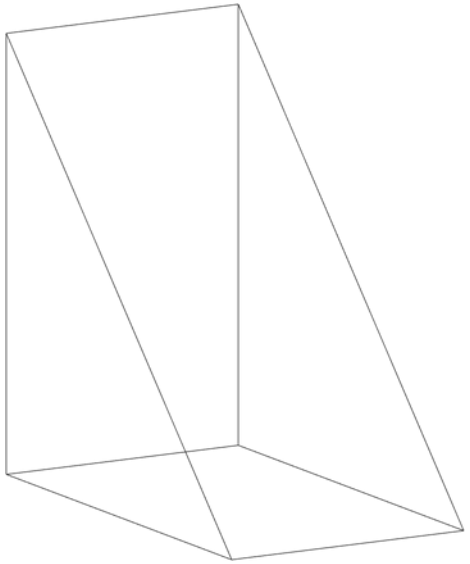
Cat image by CNX OpenStax is licensed under CC BY 4.0; changes made

1959
Hubel & Wiesel

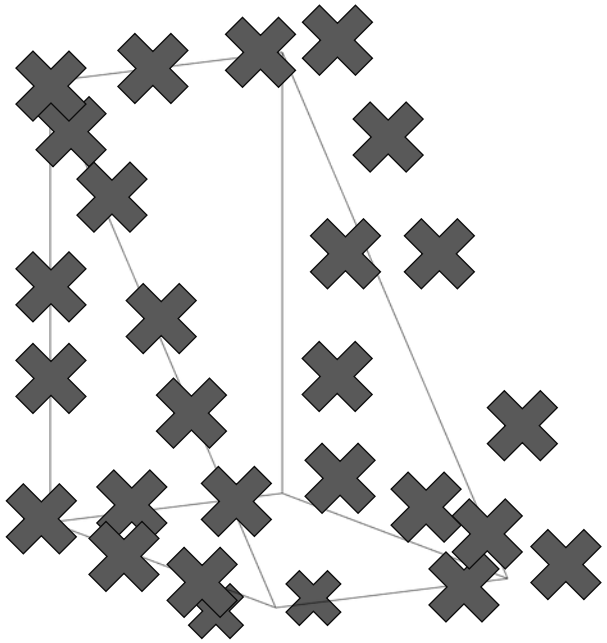
Larry Roberts, 1963



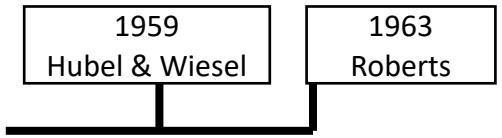
(a) Original picture



(b) Differentiated picture



(c) Feature points selected



MASSACHUSETTS INSTITUTE OF TECHNOLOGY

PROJECT MAC

Artificial Intelligence Group — July 7, 1966
Vision Memo. No. 100.

THE SUMMER VISION PROJECT

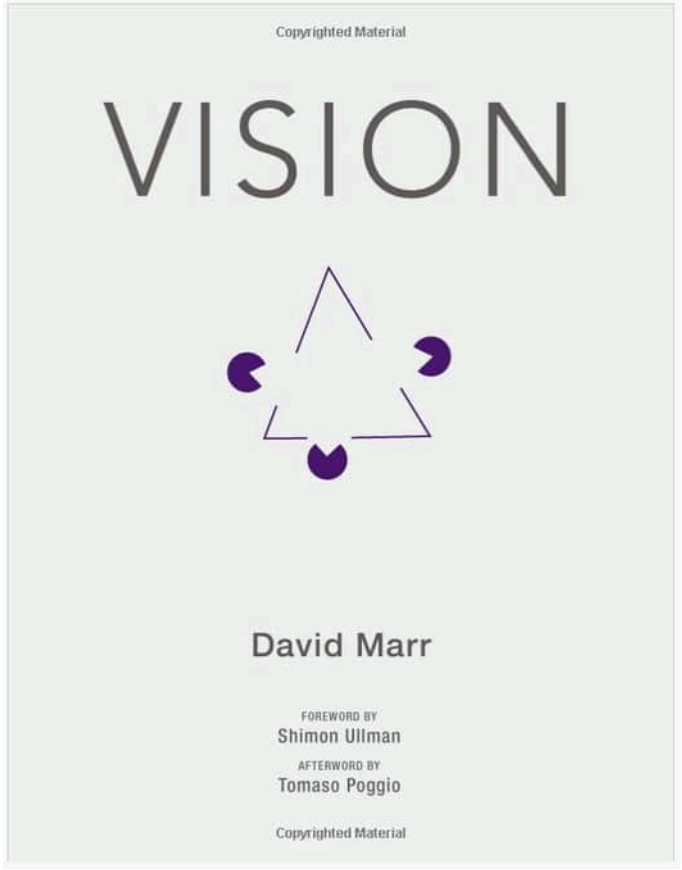
Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

1959 Hubel & Wiesel

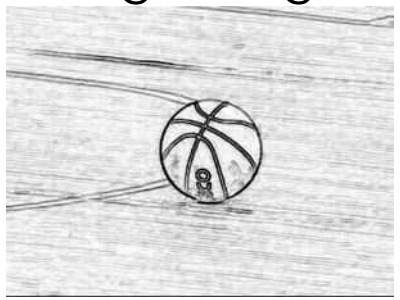
1963 Roberts

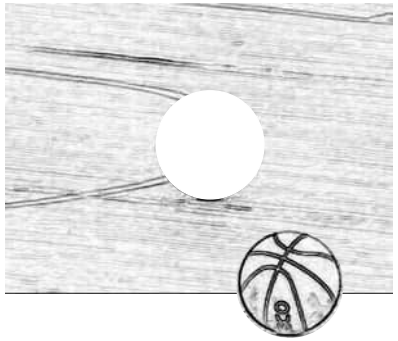
https://dspace.mit.edu/handle/1721.1/6125

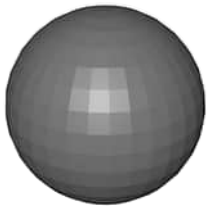




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This image is CC0 1.0 public domain

Input Image	Primal Sketch	2 ½-D Sketch	3-D Model Representation
Perceived intensities	Zero crossings, blobs, edges, bars, ends, virtual lines, groups, curves boundaries	Local surface orientation and discontinuities in depth and in surface orientation	3-D models hierarchically organized in terms of surface and volumetric primitives

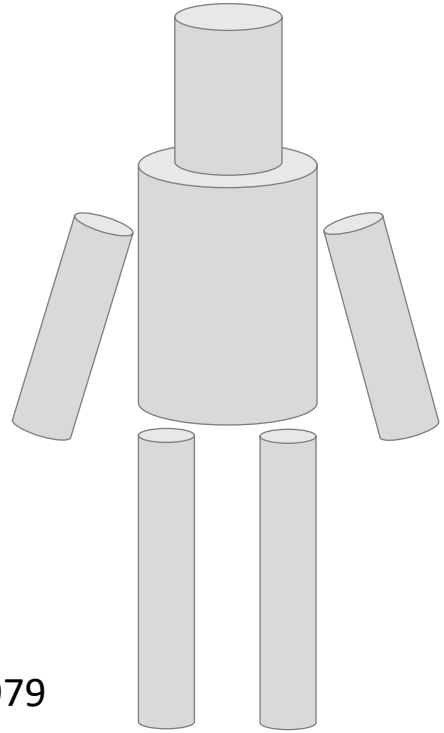
1959 Hubel & Wiesel

1963 Roberts

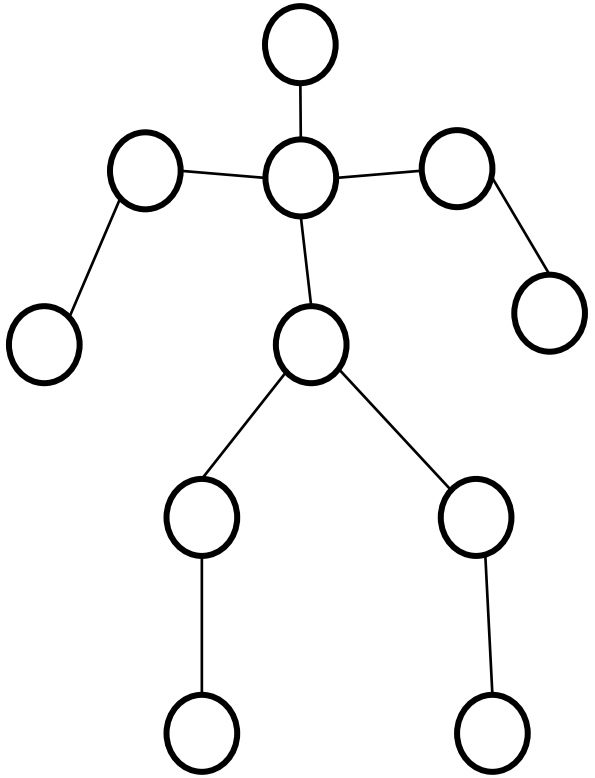
1970s David Marr

Stages of Visual Representation, David Marr, 1970s

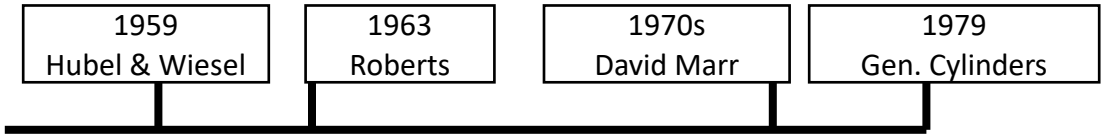
Recognition via Parts (1970s)



Generalized Cylinders,
Brooks and Binford, 1979



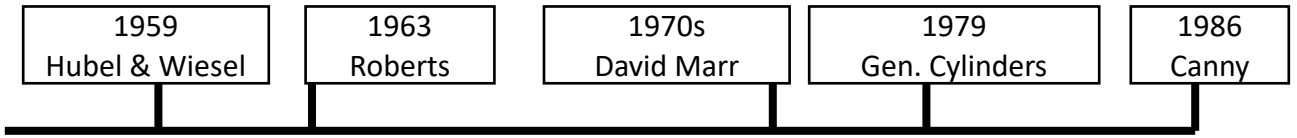
Pictorial Structures,
Fischler and Elshlager, 1973



Recognition via Edge Detection (1980s)







John Canny, 1986
David Lowe, 1987

Image is CC0 1.0 public domain

Recognition via Grouping (1990s)



1959 Hubel & Wiesel

1963 Roberts

1970s David Marr

1979 Gen. Cylinders

1986 Canny

1997 Norm. Cuts

AI Winter

Normalized Cuts, Shi and Malik, 1997

Left Image is CC BY 3.0 Middl Image is public domain Right Image is CC-BY 2.0; changes made

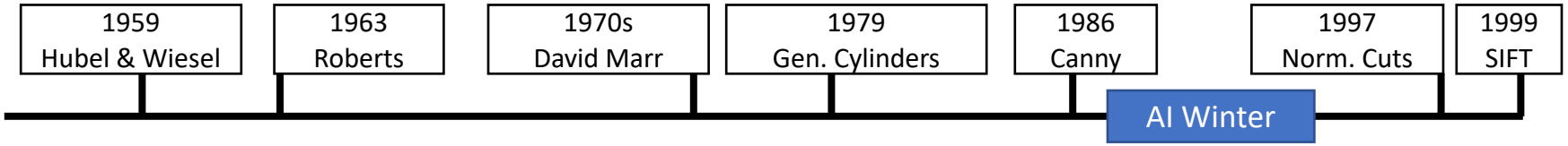
Recognition via Matching (2000s)



Image is public domain



Image is public domain



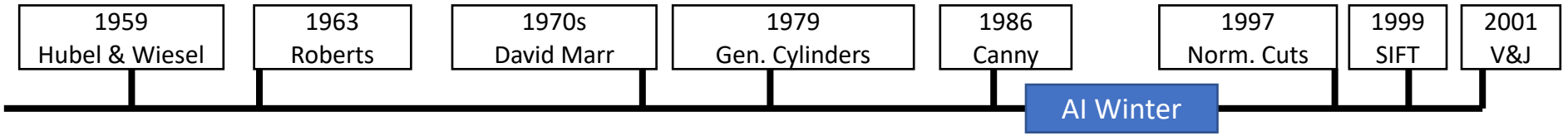
SIFT, David Lowe, 1999

Face Detection

Viola and Jones, 2001

One of the first successful applications of machine learning to vision



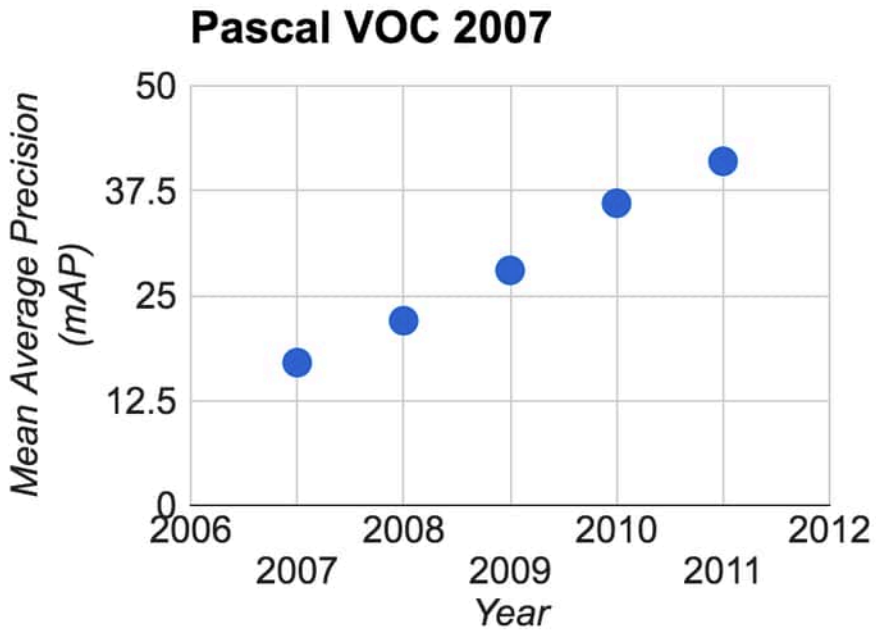


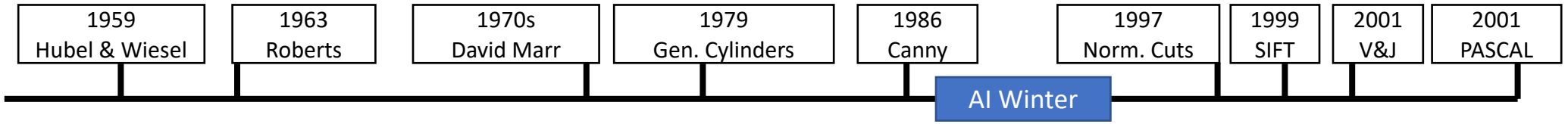
PASCAL Visual Object Challenge

Image is CC0 1.0 public domain



Image is CC0 1.0 public domain





IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:

1,000 object classes

1,431,167 images



Output:

Scale

T-shirt

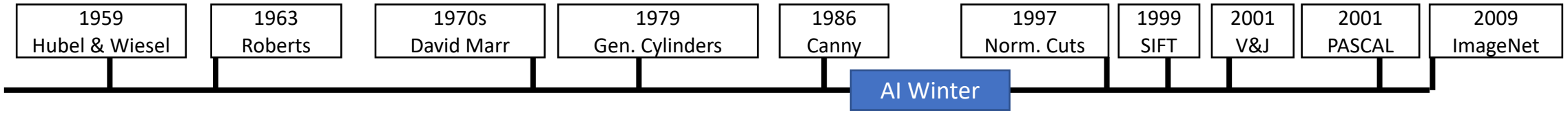
Steel drum

Drumstick

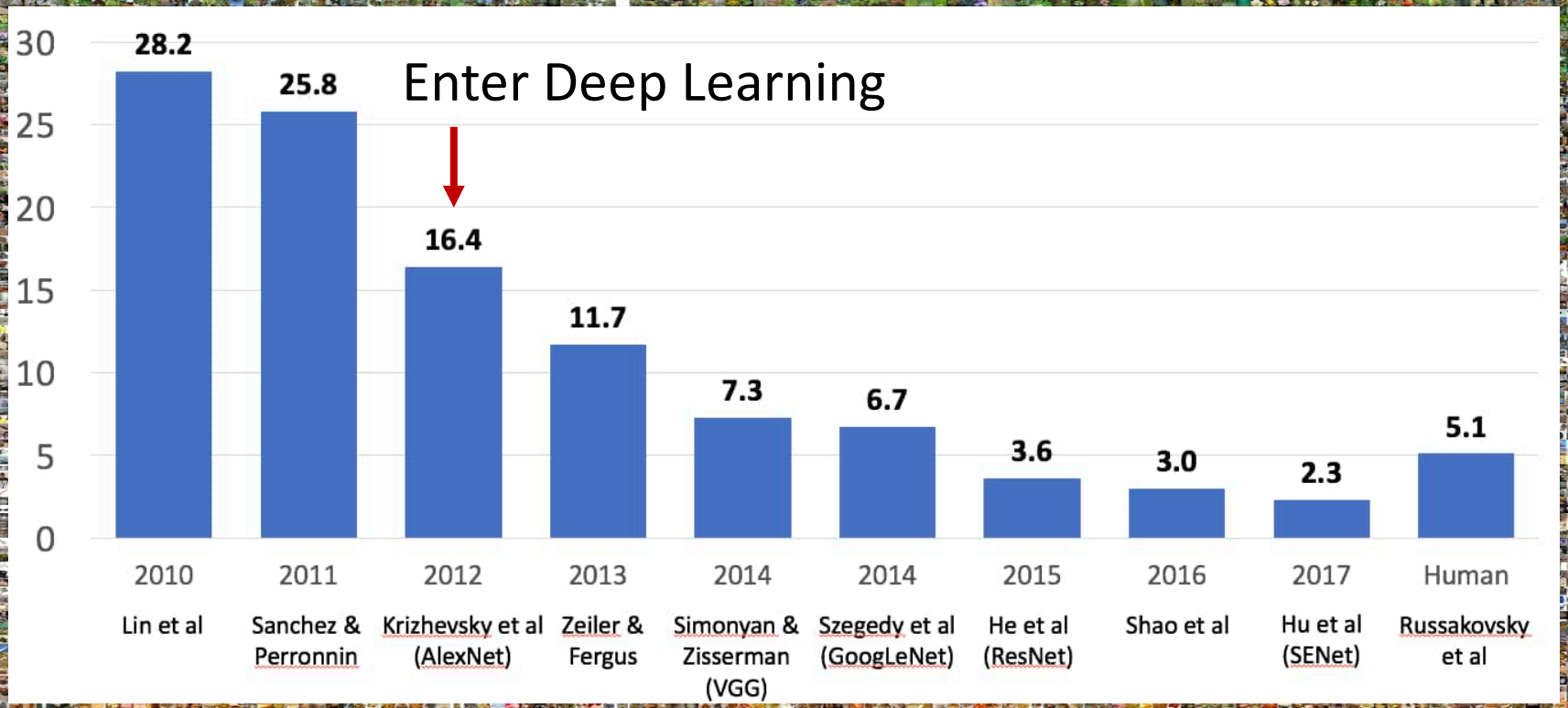
Mud turtle

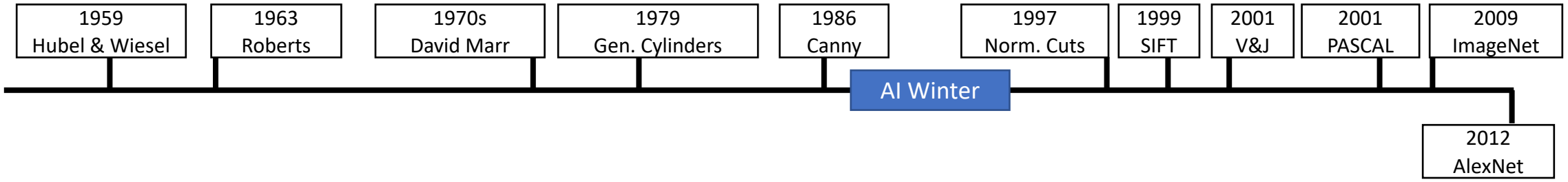
Deng et al, 2009

Russakovsky et al. IJCV 2015

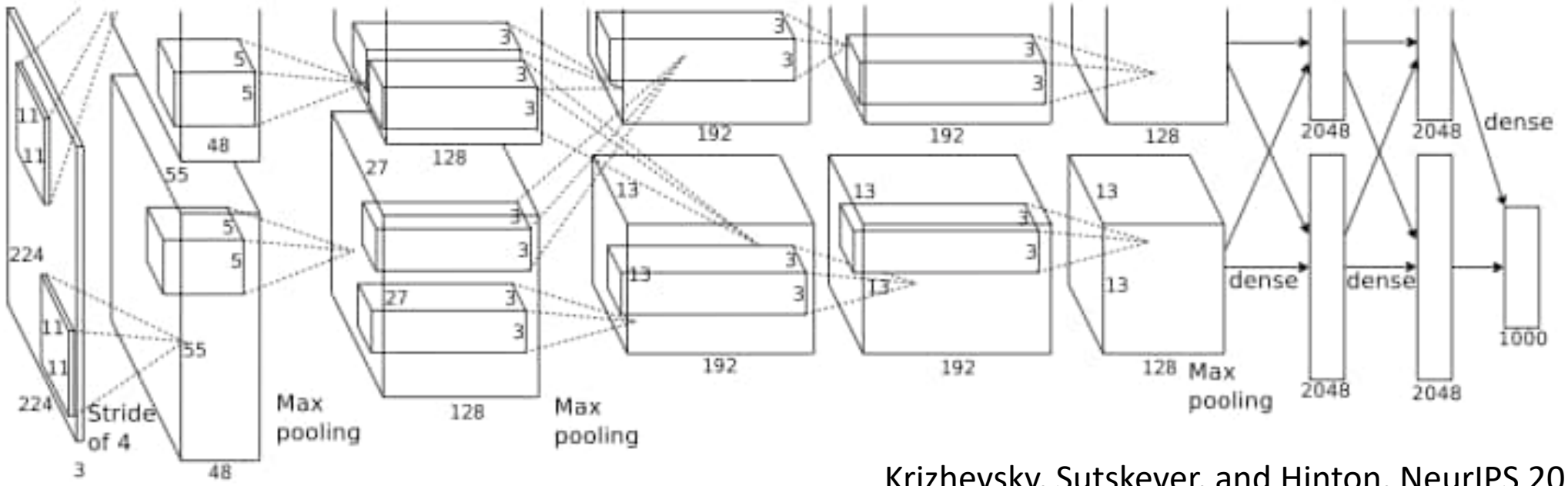


IMAGENET Large Scale Visual Recognition Challenge

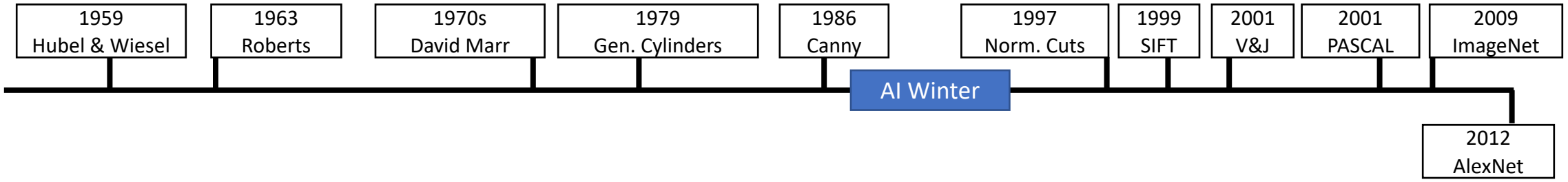




AlexNet: Deep Learning Goes Mainstream



Krizhevsky, Sutskever, and Hinton, NeurIPS 2012



Perceptron

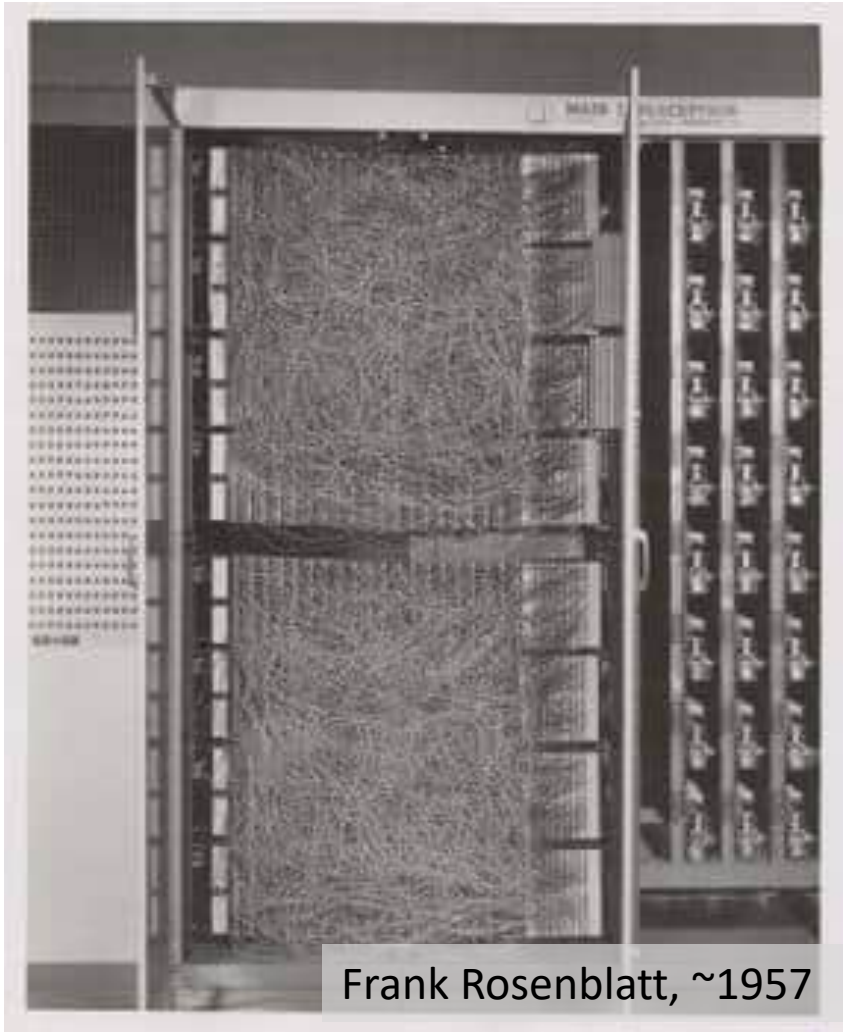
One of the earliest algorithms that could learn from data

Implemented in hardware! Weights stored in potentiometers, updated with electric motors during learning

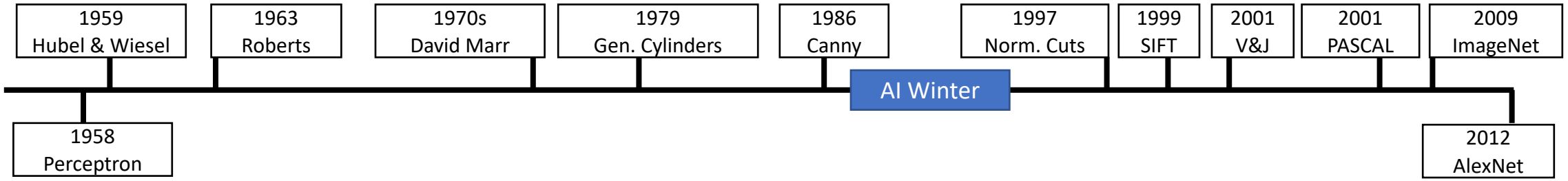
Connected to a camera that used 20x20 cadmium sulfide photocells to make a 400-pixel image

Could learn to recognize letters of the alphabet

Today we would recognize it as a **linear classifier**

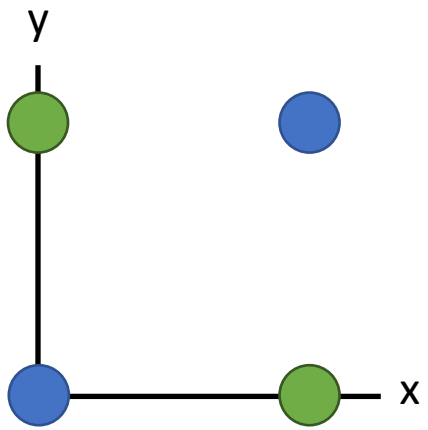


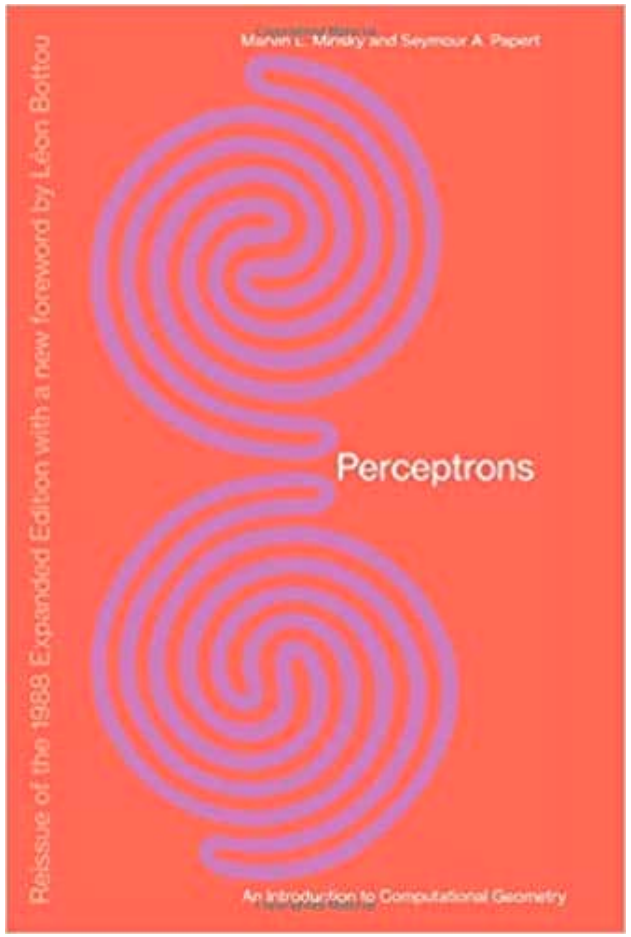
Frank Rosenblatt, ~1957



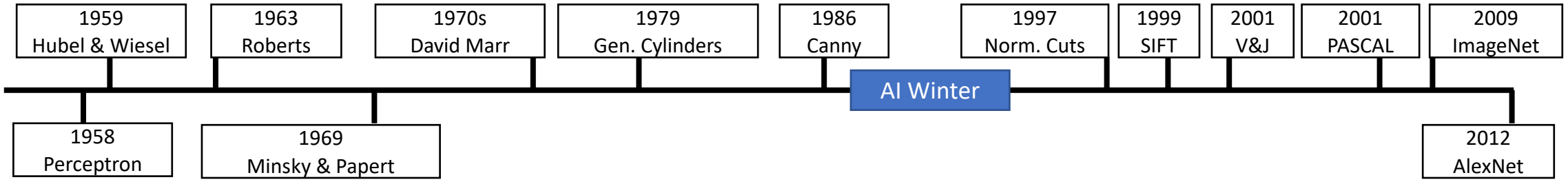
Minsky and Papert, 1969

X	Y	F(x,y)
0	0	0
0	1	1
1	0	1
1	1	0





Showed that Perceptrons could not learn the XOR function
Caused a lot of disillusionment in the field

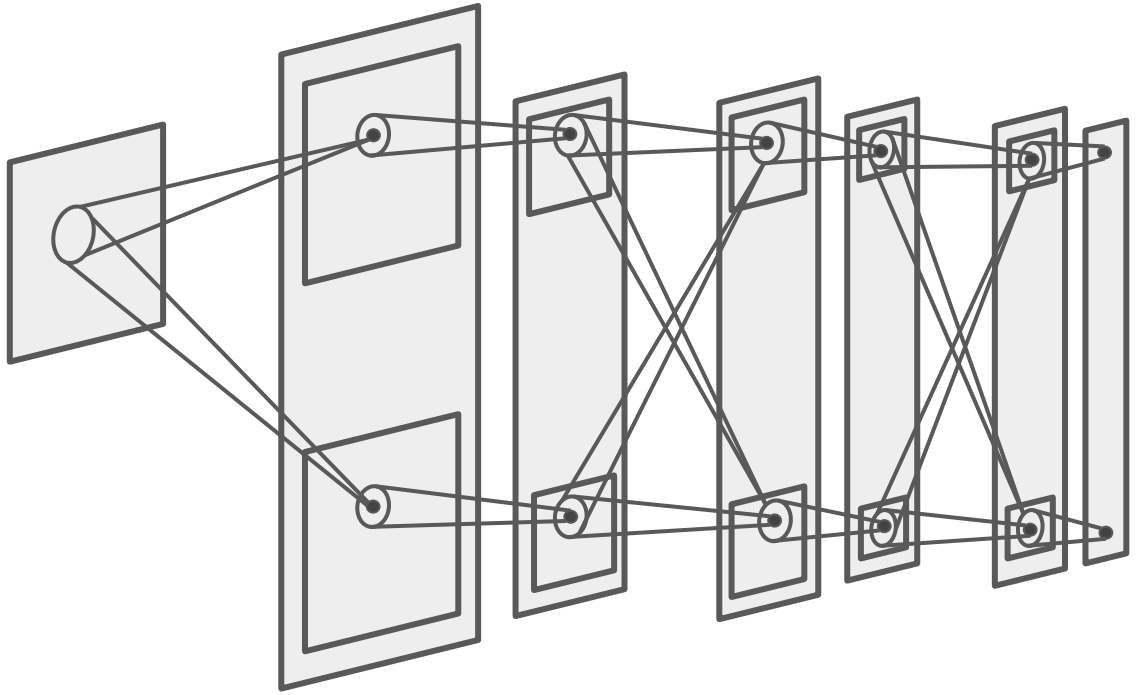


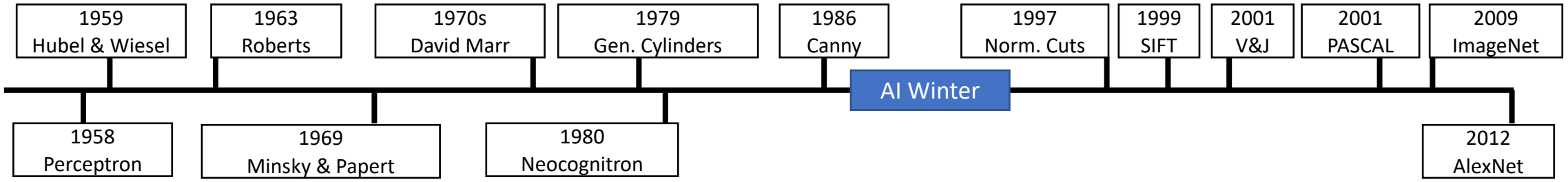
Neocognitron: Fukushima, 1980

Computational model the visual system, directly inspired by Hubel and Wiesel's hierarchy of complex and simple cells

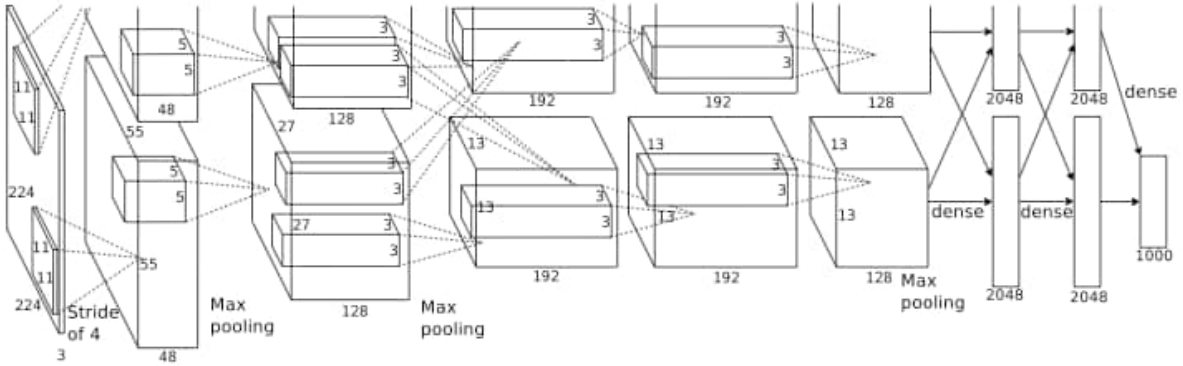
Interleaved simple cells (convolution) and complex cells (pooling)

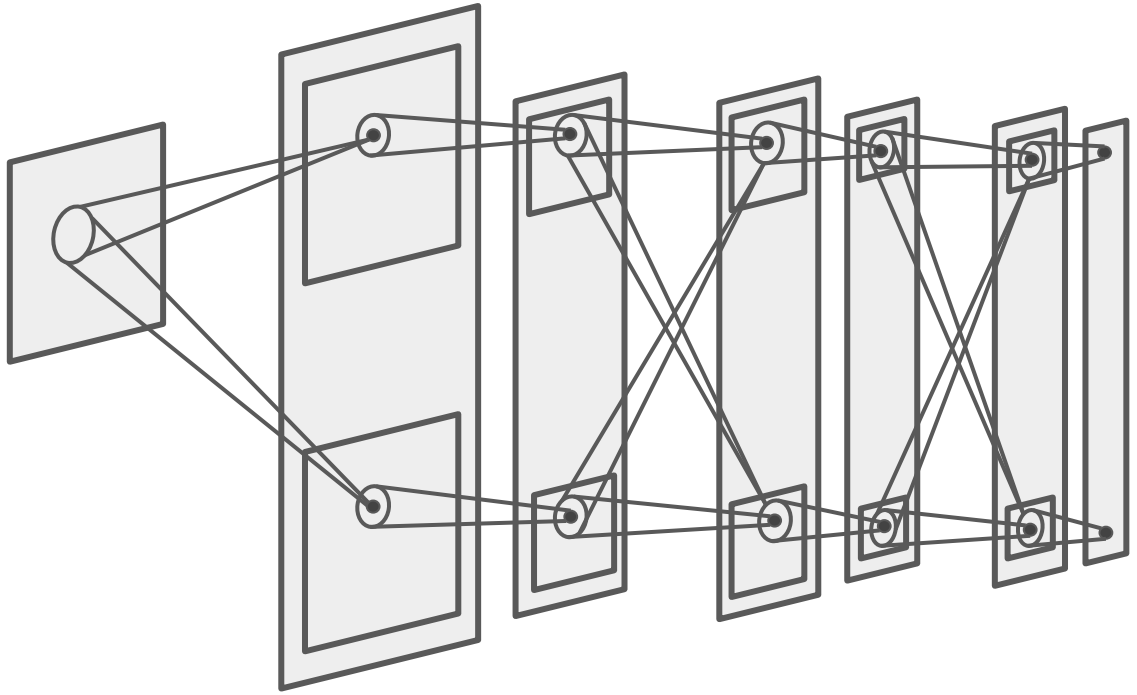
No practical training algorithm



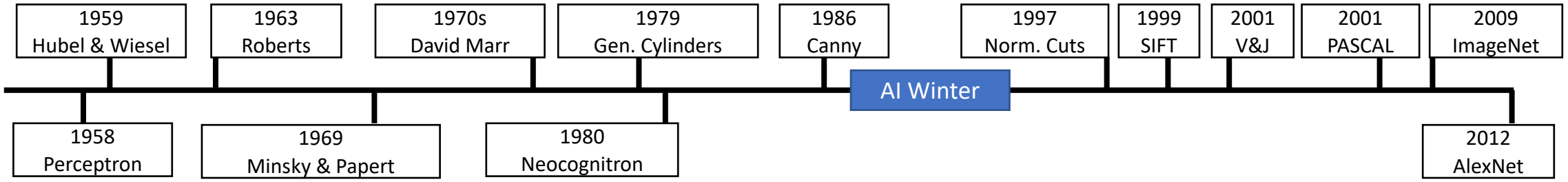


Neocognitron: Fukushima, 1980





Looks a lot like AlexNet
more than 32 years later!



Backprop: Rumelhart, Hinton, and Williams, 1986

Introduced backpropagation for computing gradients in neural networks

Successfully trained perceptrons with multiple layers

$$\frac{\partial E_p}{\partial w_{ji}} = \frac{\partial E_p}{\partial o_{pj}} \frac{\partial o_{pj}}{\partial w_{ji}}$$

recognizable math

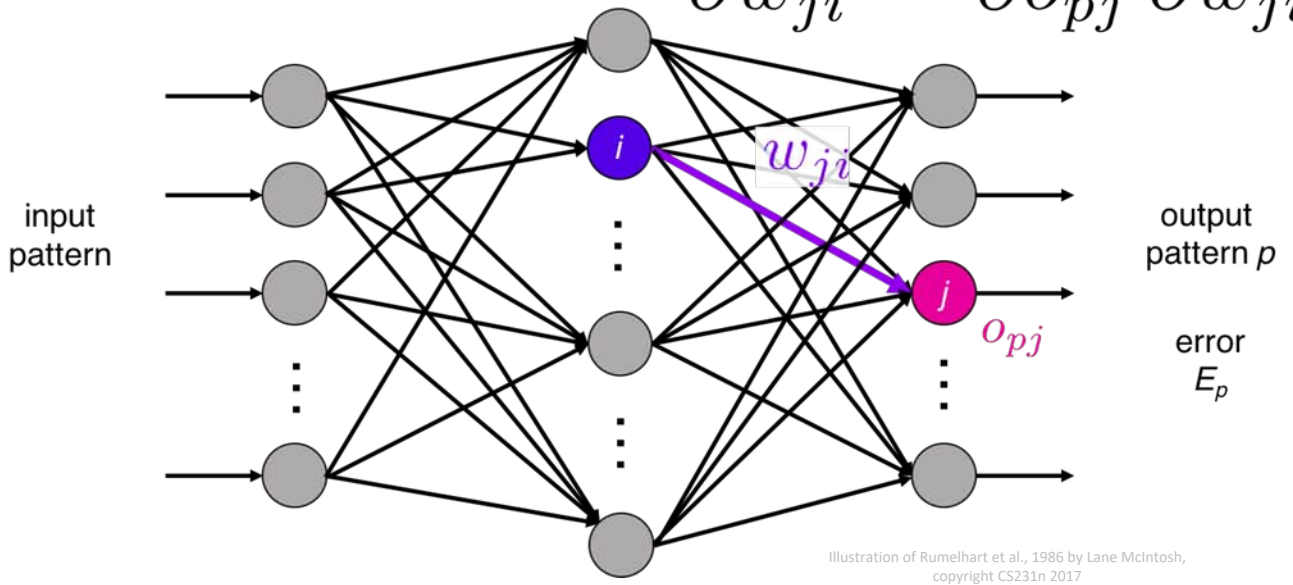
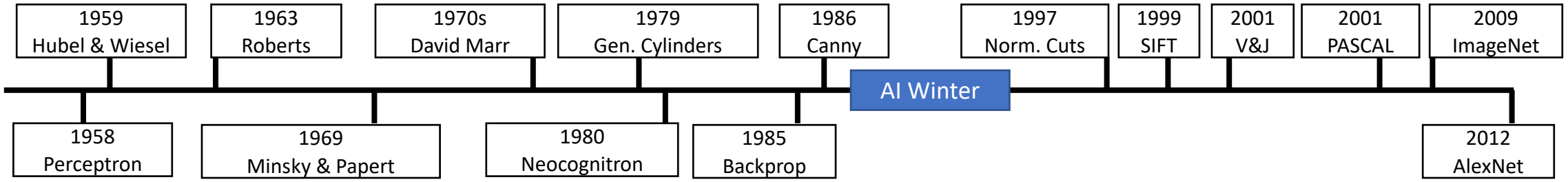
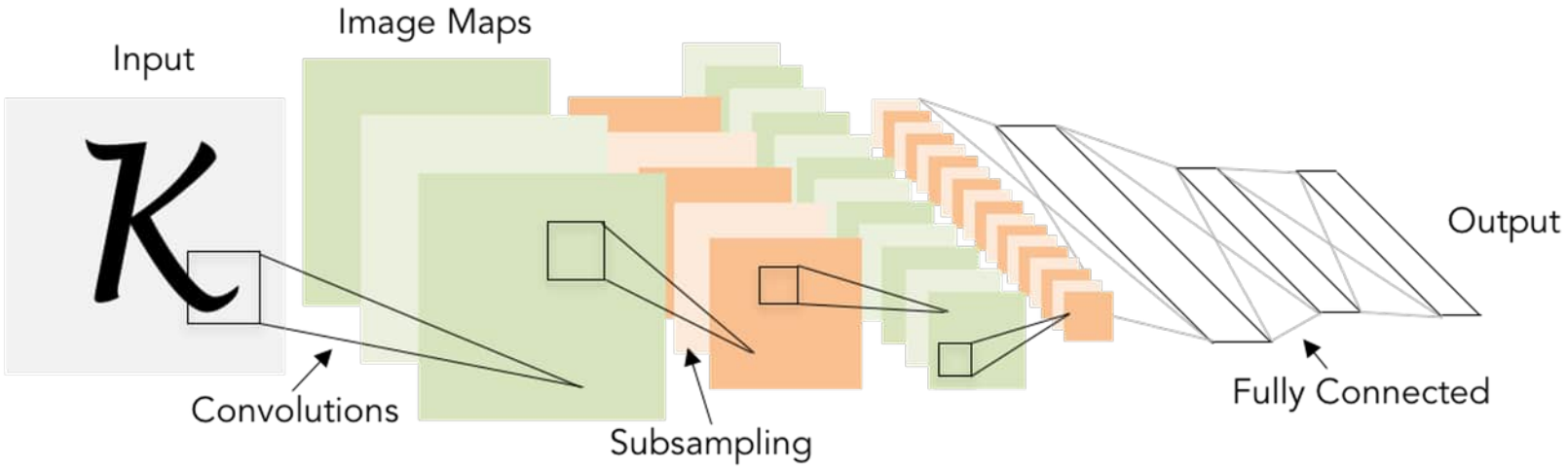


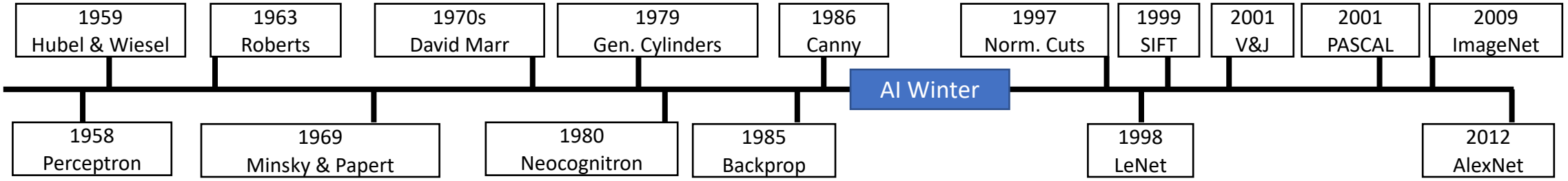
Illustration of Rumelhart et al., 1986 by Lane McIntosh, copyright CS231n 2017



Convolutional Networks: LeCun et al, 1998



Applied backprop algorithm to a Neocognitron-like architecture
Learned to recognize handwritten digits
Was deployed in a commercial system by NEC, processed handwritten checks
Very similar to our modern convolutional networks!

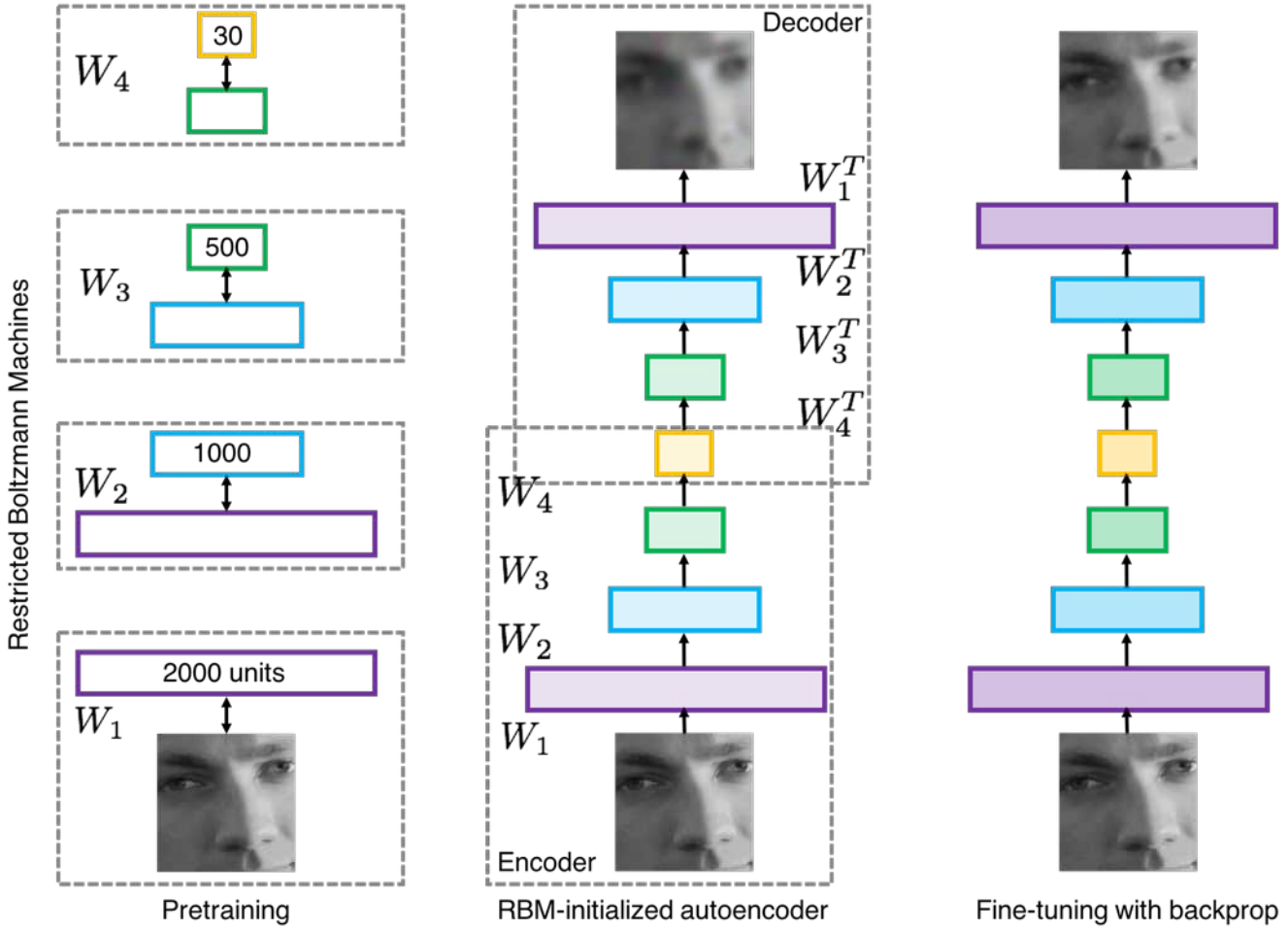


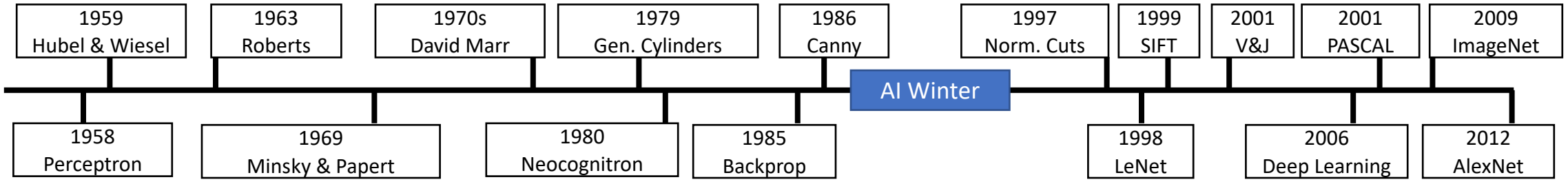
2000s: “Deep Learning”

People tried to train neural networks that were deeper and deeper

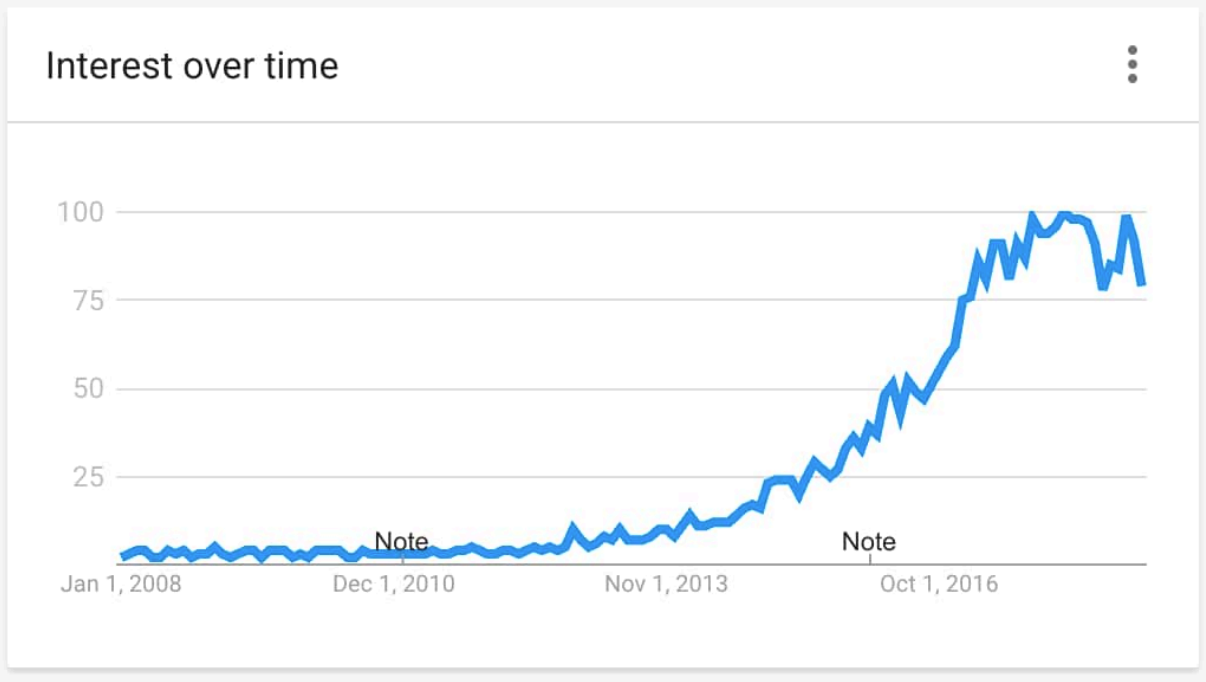
Not a mainstream research topic at this time

Hinton and Salakhutdinov, 2006
Bengio et al, 2007
Lee et al, 2009
Glorot and Bengio, 2010

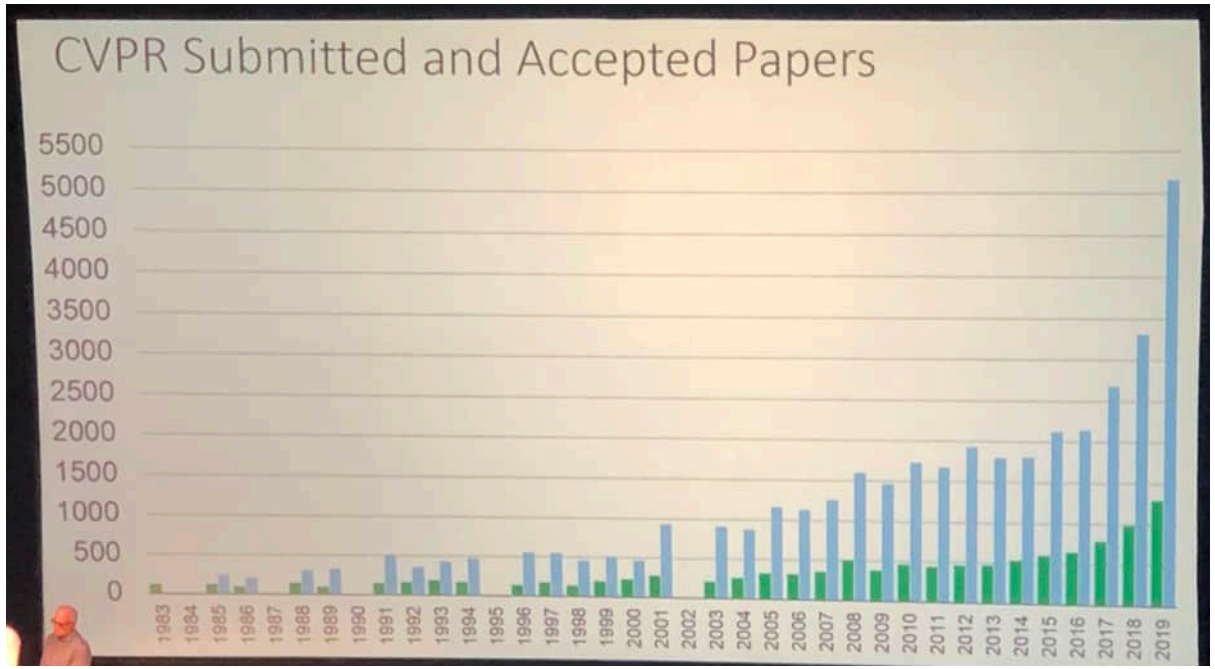




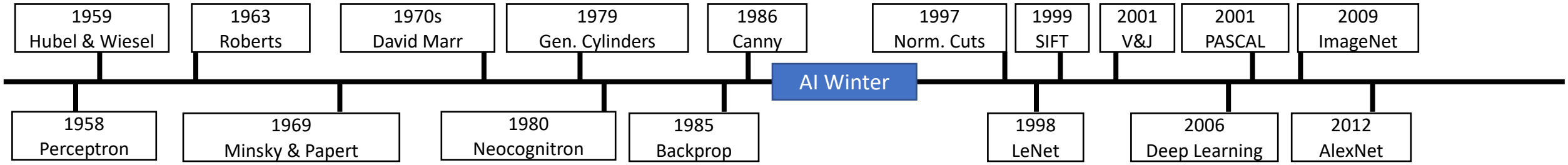
2012 to Present: Deep Learning Explosion



Google Trends: "Deep Learning"



Publications at top Computer Vision conference



2012 to Present: ConvNets are everywhere

Image Classification

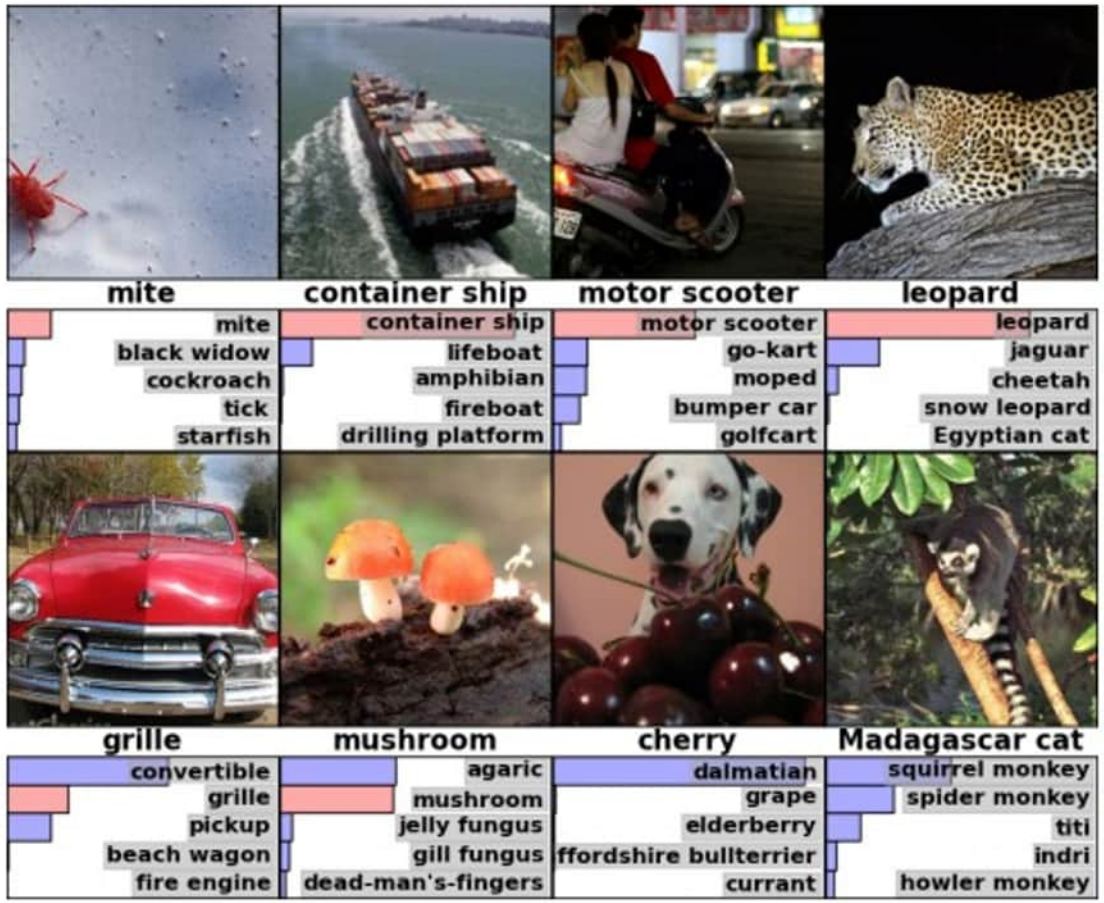


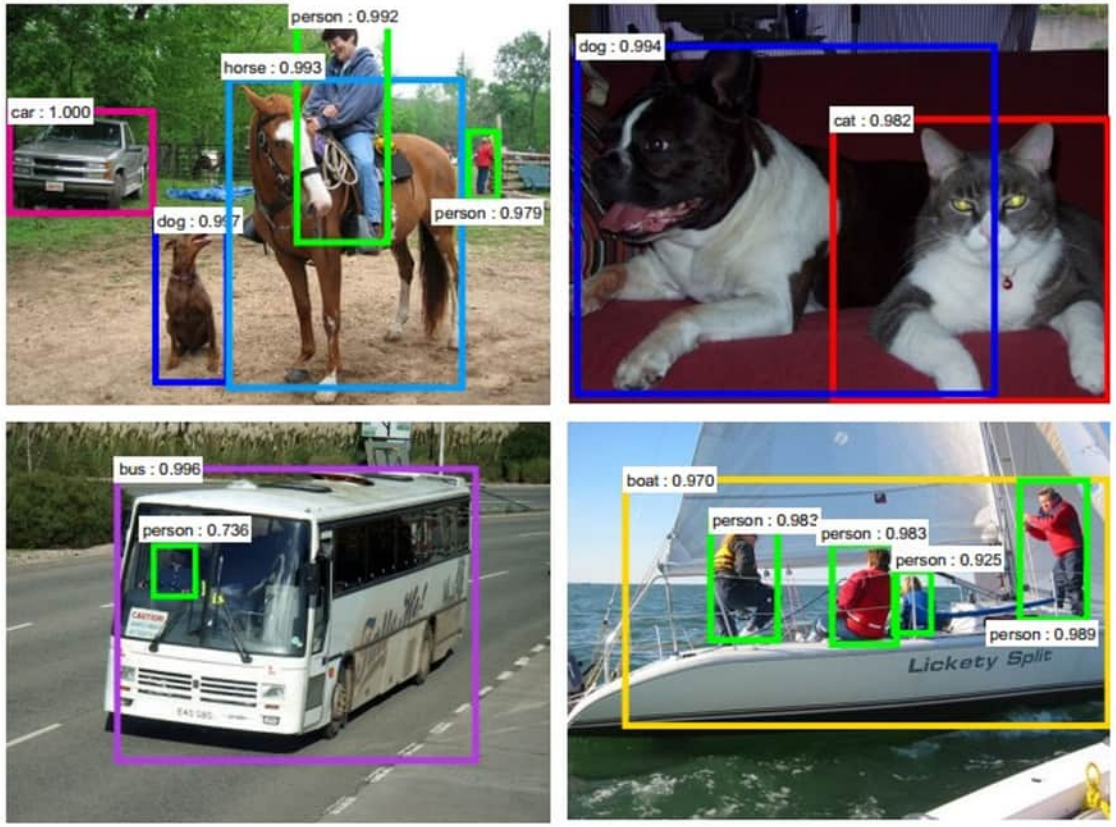
Image Retrieval



Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

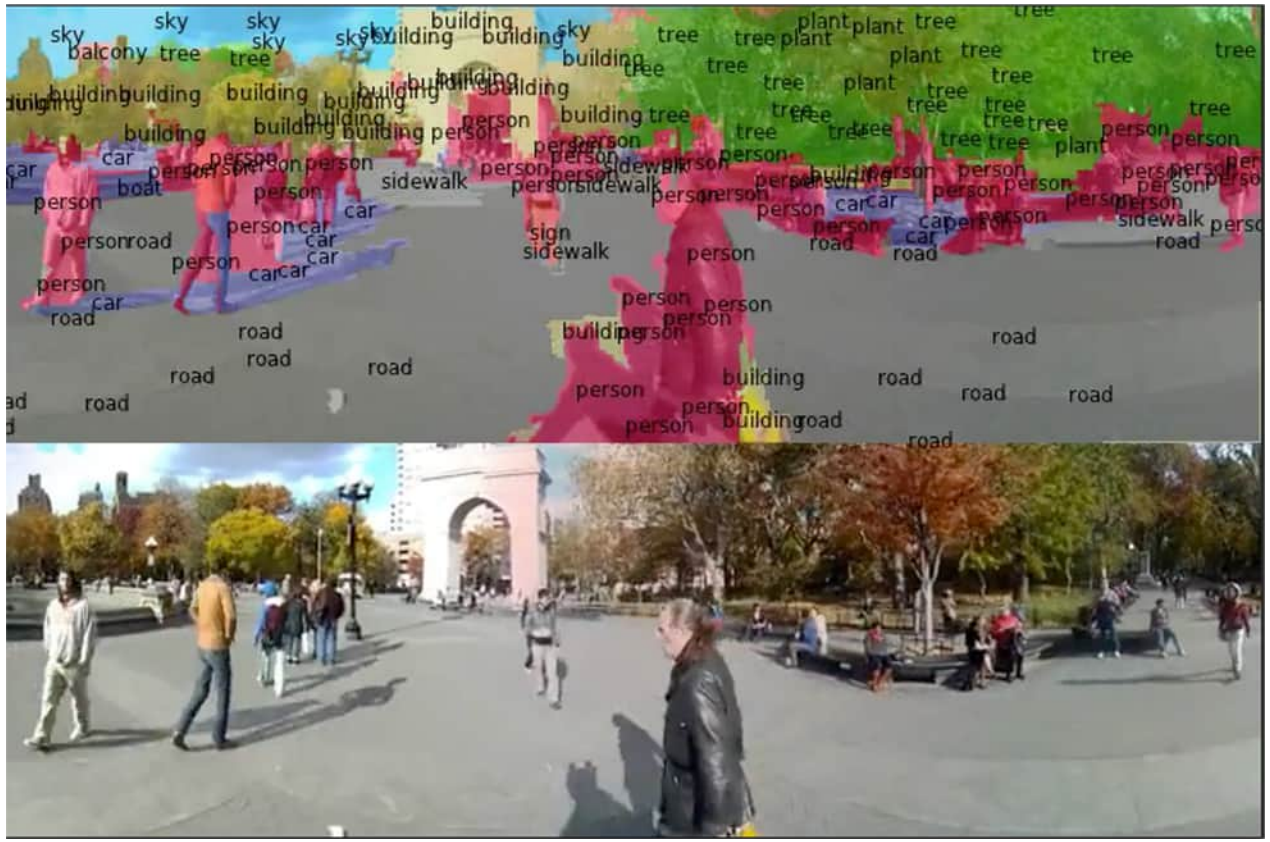
2012 to Present: ConvNets are everywhere

Object Detection



Ren, He, Girshick, and Sun, 2015

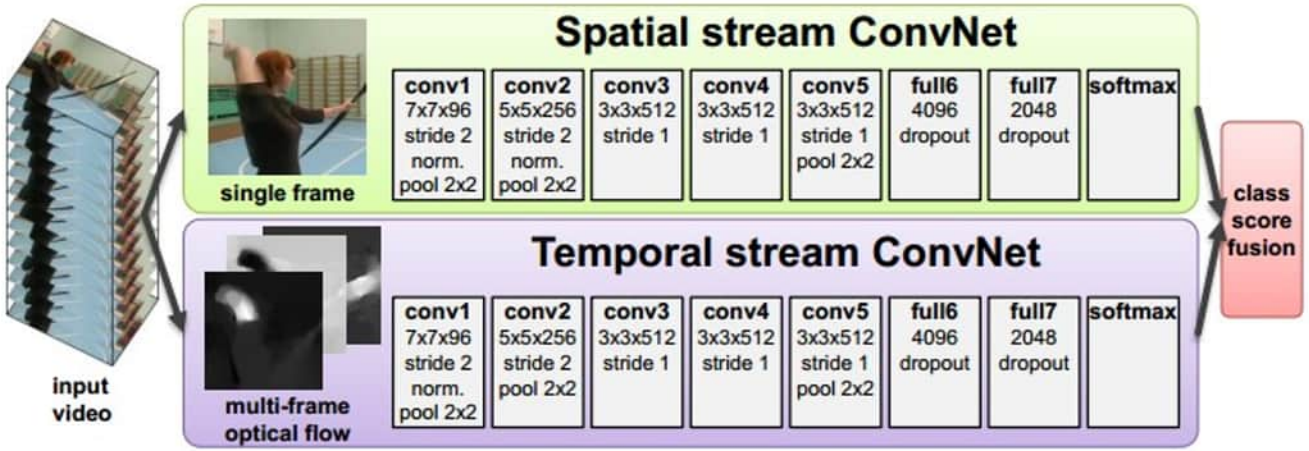
Image Segmentation



Fabaret et al, 2012

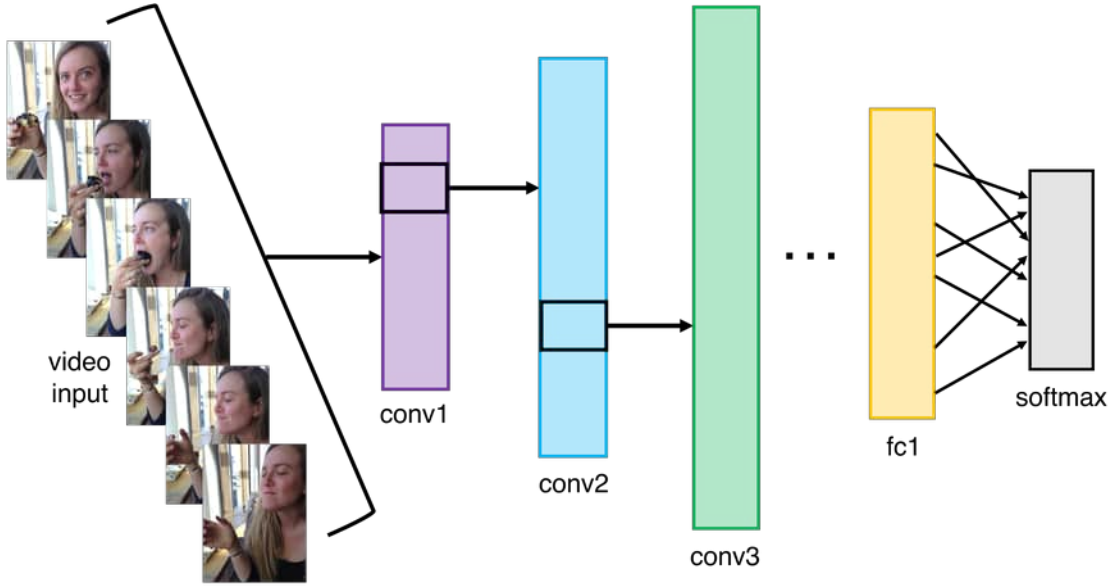
2012 to Present: ConvNets are everywhere

Video Classification



Simonyan et al, 2014

Activity Recognition

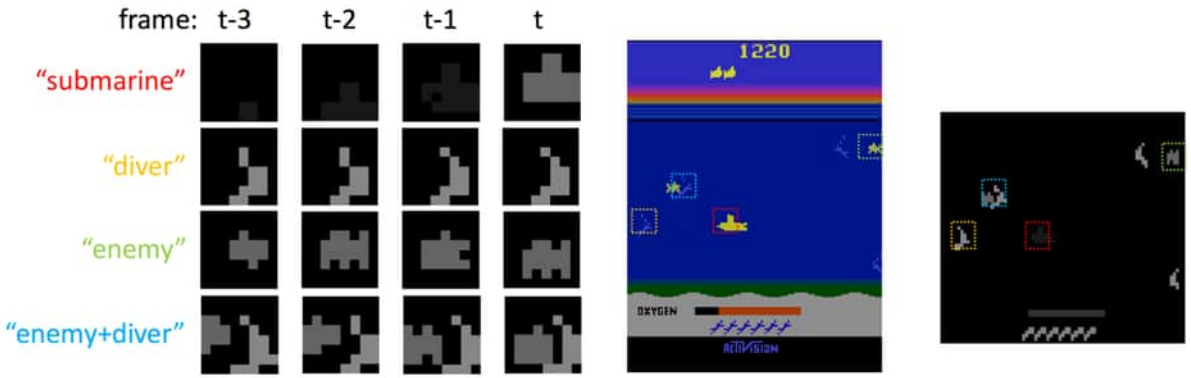


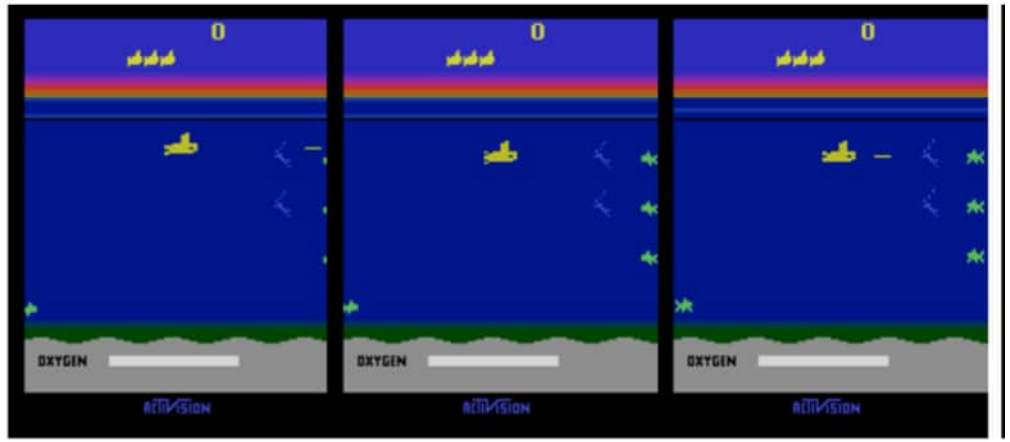
2012 to Present: ConvNets are everywhere

Pose Recognition (Toshev and Szegedy, 2014)



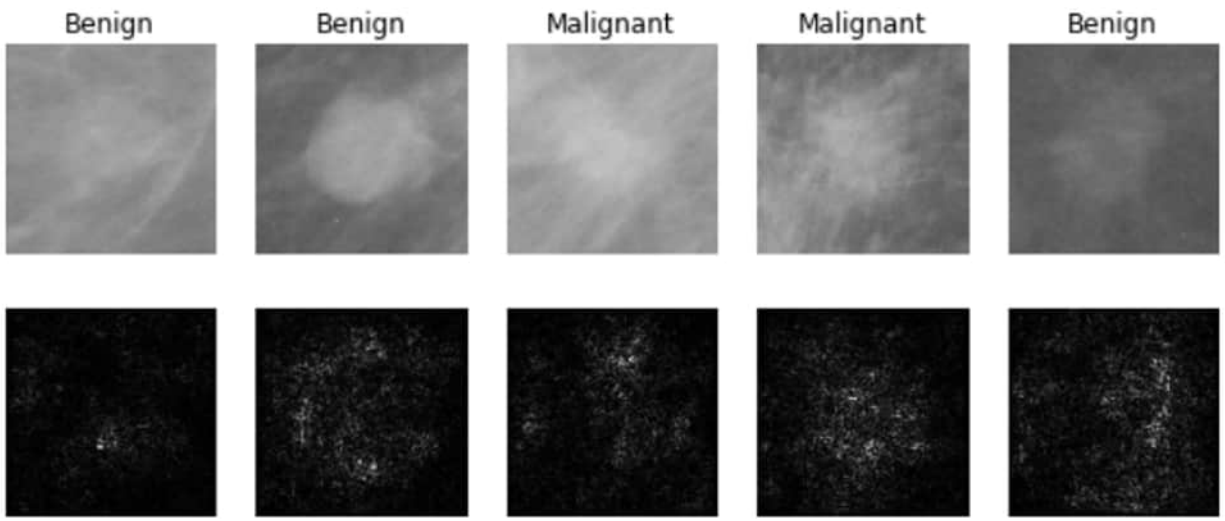
Playing Atari games (Guo et al, 2014)





2012 to Present: ConvNets are everywhere

Medical Imaging



Levy et al, 2016 Figure reproduced with permission

Galaxy Classification



Dieleman et al, 2014

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Whale recognition



Kaggle Challenge

This image by Christin Khan is in the public domain and originally came from the U.S. NOAA.

2012 to Present: ConvNets are everywhere

Image Captioning

Vinyals et al, 2015
Karpathy and Fei-Fei, 2015



A white teddy bear sitting in the grass



A man in a baseball uniform throwing a ball



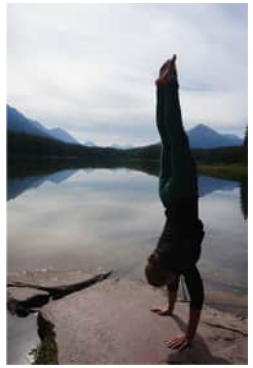
A woman is holding a cat in her hand



A man riding a wave on top of a surfboard



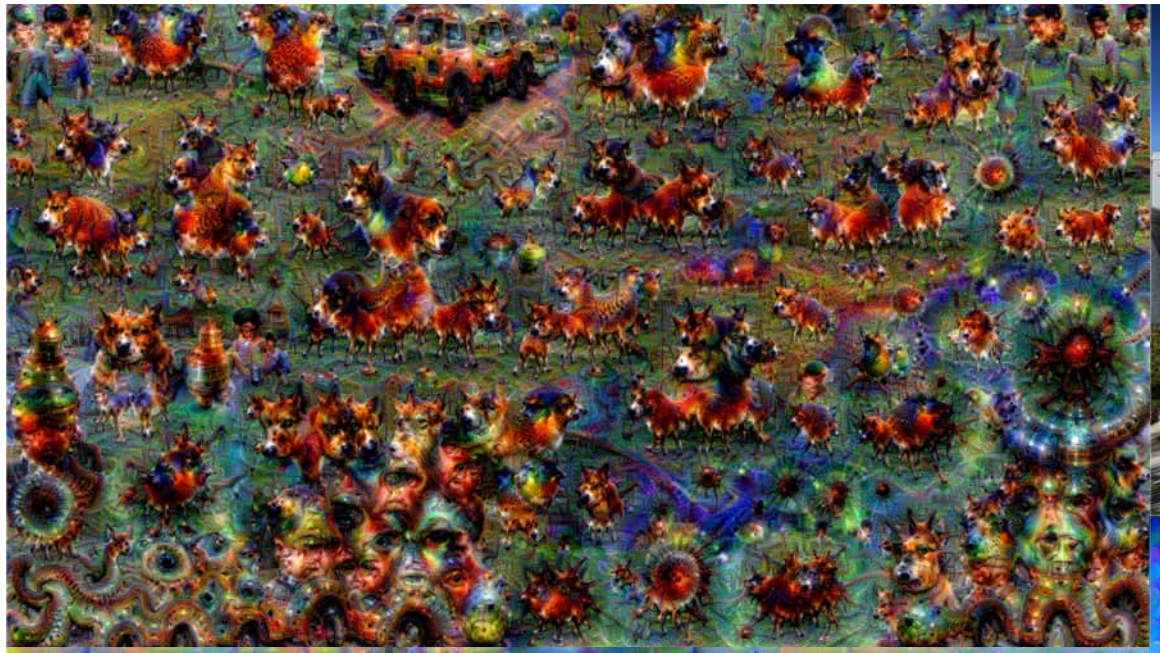
A cat sitting on a suitcase on the floor

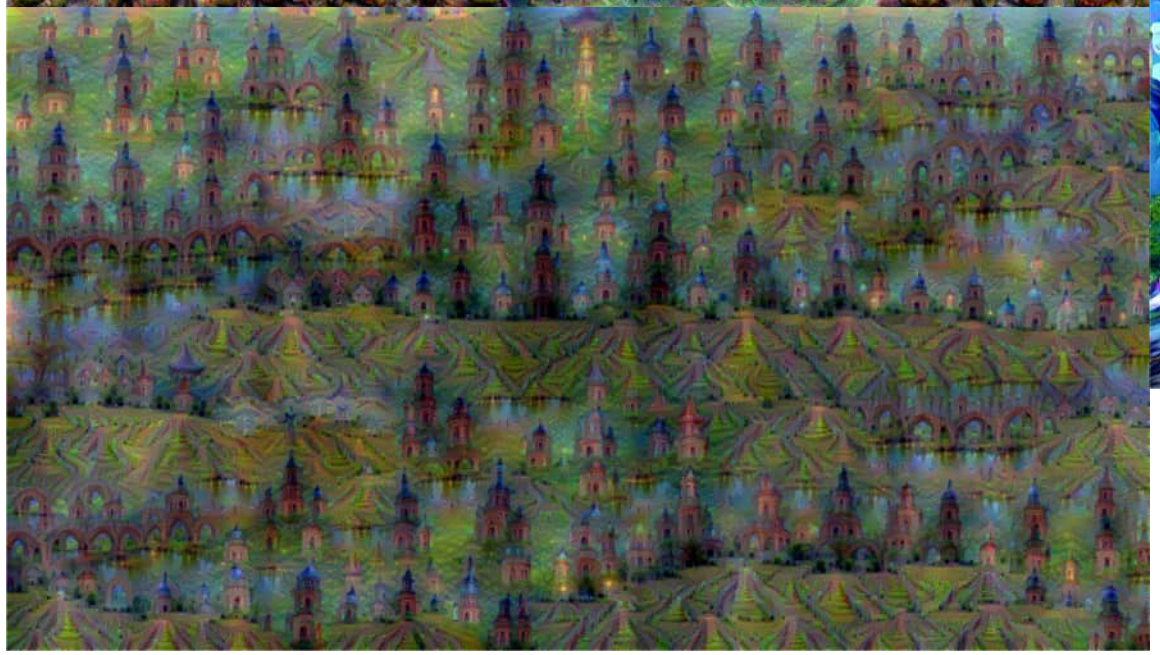


A woman standing on a beach holding a surfboard

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https://pixabay.com/en/teddy-plush-bears-cute-teddy-bear-1623436/
https://pixabay.com/en/surf-wave-summer-sport-litoral-1668716/
https://pixabay.com/en/woman-female-model-portrait-adult-983967/
https://pixabay.com/en/handstand-lake-meditation-496008/
https://pixabay.com/en/baseball-player-shortstop-infield-1045263/

Captions generated by Justin Johnson using Neuraltalk2













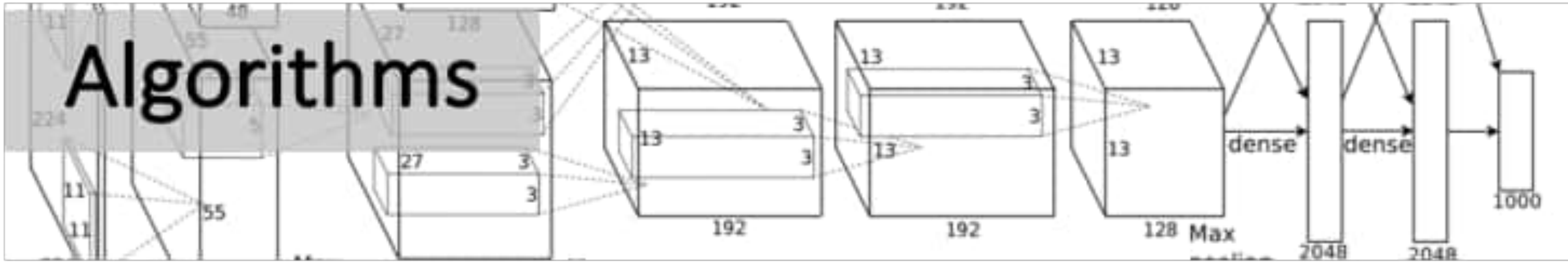




Figures copyright Justin Johnson, 2015. Reproduced with permission. Generated using the Inceptionism approach from a blog post by Google Research.

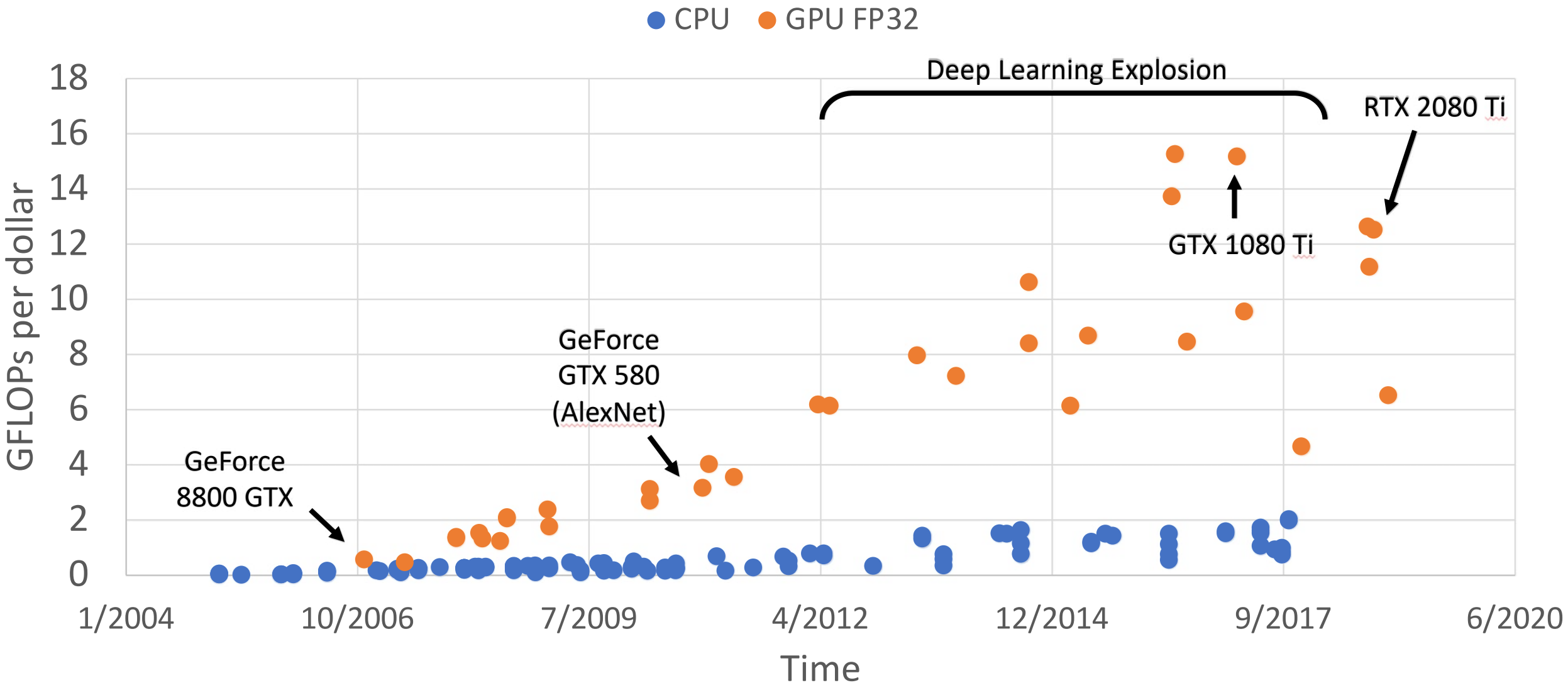
Original image is CC0 public domain
Starry Night and Tree Roots by Van Gogh are in the public domain
Bokeh image is in the public domain
Stylized images copyright Justin Johnson, 2017;
reproduced with permission

Mordvinsev et al, 2015
Gatys et al, 2016

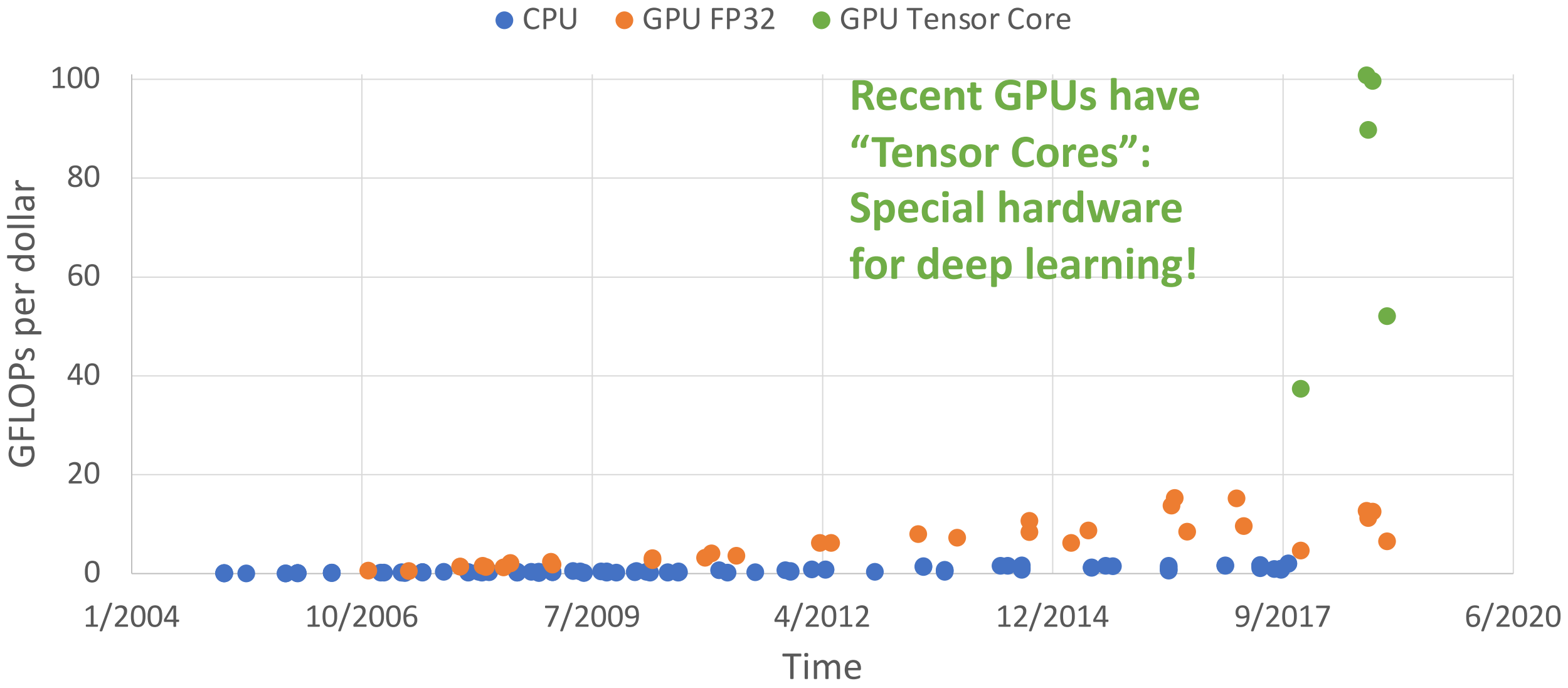

Algorithms
13
13
3
13
3
192
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3
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3
192
13
13
128
Max
dense
dense
2048
2048
1000


Data
Computation
GTX
GTX

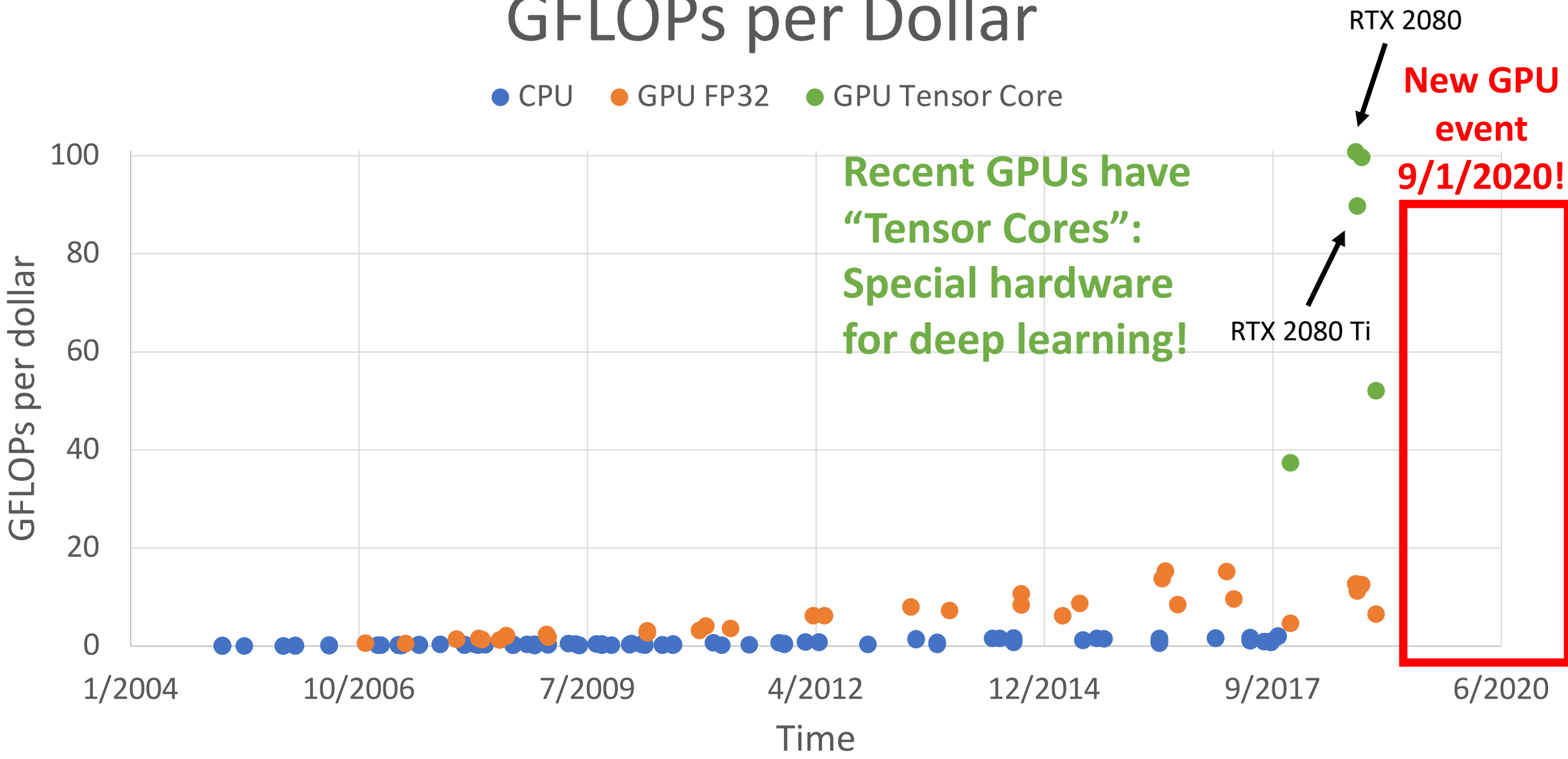
GFLOPs per Dollar



GFLOPs per Dollar



GFLOPs per Dollar

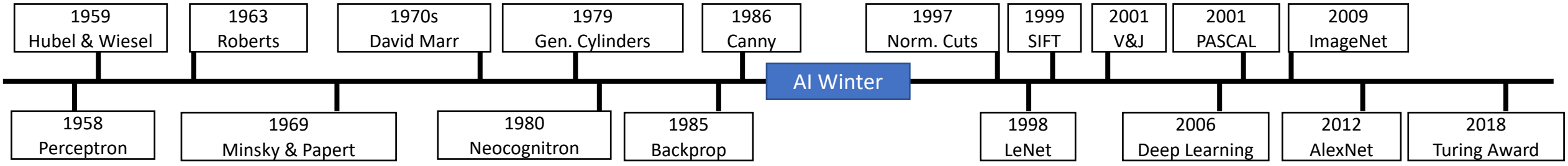


2018 Turing Award


Yoshua Bengio


Geoffrey Hinton


Yann LeCun



Despite our success, computer vision still has a long way to go…



This image is copyright-free United States government work

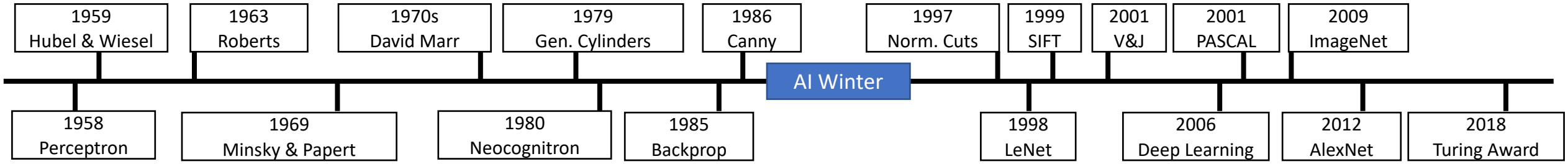
Example credit:
Andrej Karpathy



Computer Vision Technology Can Better Our Lives

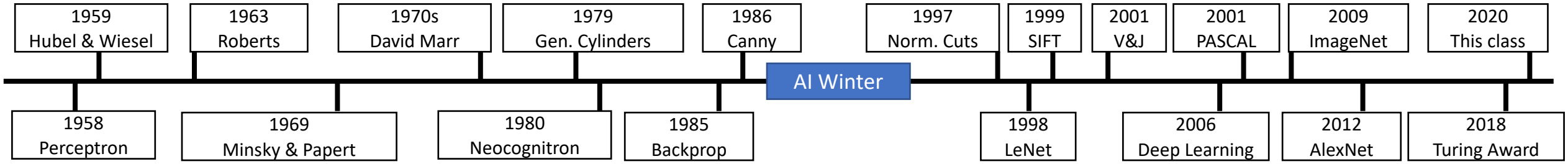
Today's Agenda

- A brief history of computer vision and deep learning
- Course overview and logistics



Today's Agenda

- A brief history of computer vision and deep learning
- Course overview and logistics



Course Staff

Instructor



Justin Johnson
Assistant Professor, CSE

Graduate Student Instructors



Yunseok Jang



Mohamed El Banani



Danish Syed

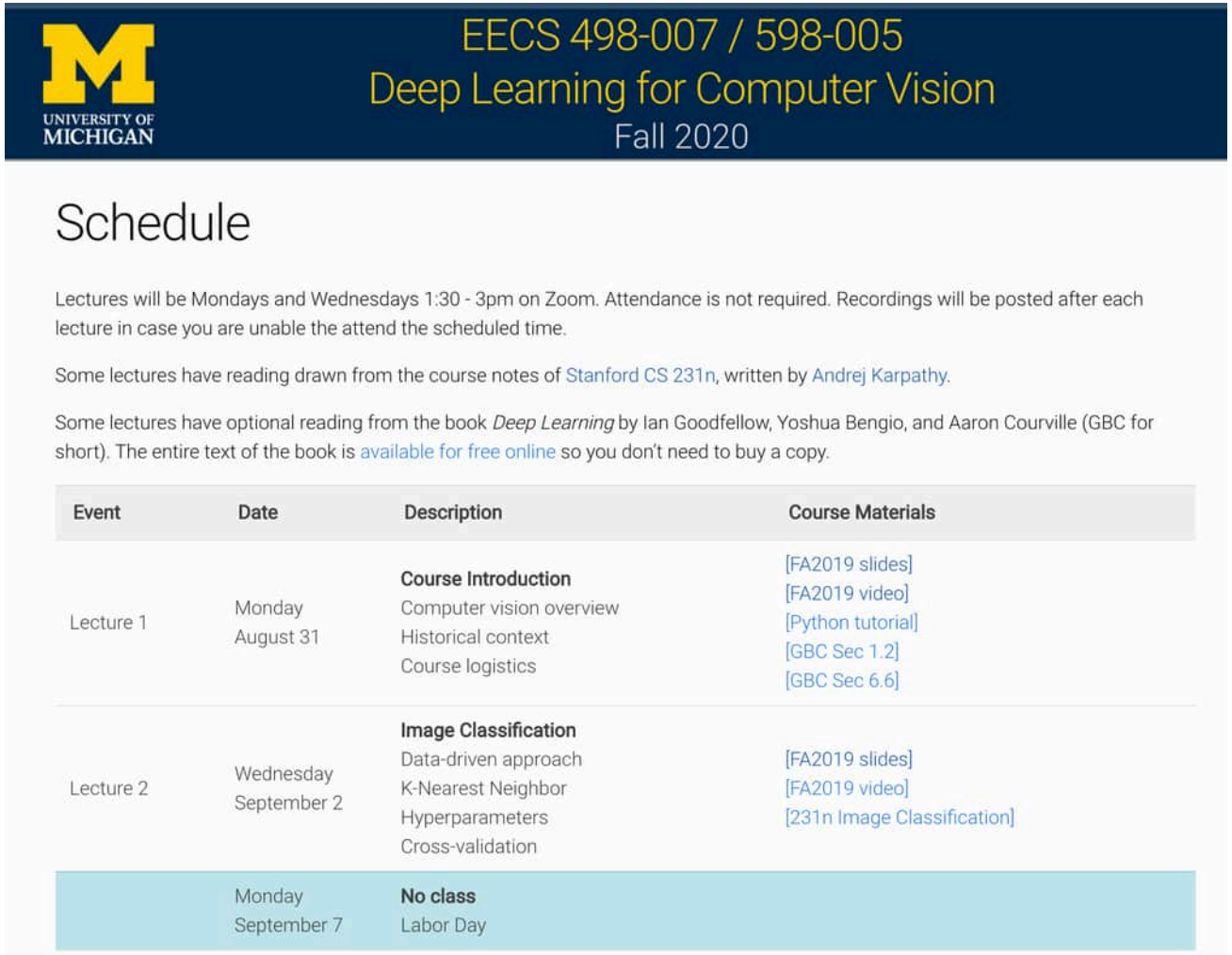


Yashmeet Gambhir

How to contact us

- Course Website: https://web.eecs.umich.edu/~justincj/teaching/eecs498/
  - Syllabus, schedule, assignments, slides, lecture videos, etc
- Piazza: https://piazza.com/class/ke3a8m6u5wx647
  - (Almost) all questions about the course should go here!
  - We will also use Piazza to communicate with you
  - Use private questions if you want to post code
- EECS Autograder:
  - For turning in homework assignments
  - Still working out details, will update soon
- Google Calendar: For office hours (starting next week)
- Email: Only for sensitive, confidential issues

Course Website: Check the Schedule!



UNIVERSITY OF MICHIGAN

EECS 498-007 / 598-005
Deep Learning for Computer Vision
Fall 2020

Schedule

Lectures will be Mondays and Wednesdays 1:30 - 3pm on Zoom. Attendance is not required. Recordings will be posted after each lecture in case you are unable the attend the scheduled time.

Some lectures have reading drawn from the course notes of Stanford CS 231n, written by Andrej Karpathy.

Some lectures have optional reading from the book *Deep Learning* by Ian Goodfellow, Yoshua Bengio, and Aaron Courville (GBC for short). The entire text of the book is available for free online so you don't need to buy a copy.

Event	Date	Description	Course Materials
Lecture 1	Monday August 31	**Course Introduction** Computer vision overview Historical context Course logistics	[FA2019 slides] [FA2019 video] [Python tutorial] [GBC Sec 1.2] [GBC Sec 6.6]
Lecture 2	Wednesday September 2	**Image Classification** Data-driven approach K-Nearest Neighbor Hyperparameters Cross-validation	[FA2019 slides] [FA2019 video] [231n Image Classification]
	Monday September 7	**No class** Labor Day	

https://web.eecs.umich.edu/~justincj/teaching/eecs498/FA2020/schedule.html

Piazza Etiquette

- Post only short snippets of code (< 20-30 lines)
- Ask a specific, concrete question
- Explain what you have tried so far, and what happened
- See StackOverflow guide on asking good questions: https://stackoverflow.com/help/how-to-ask

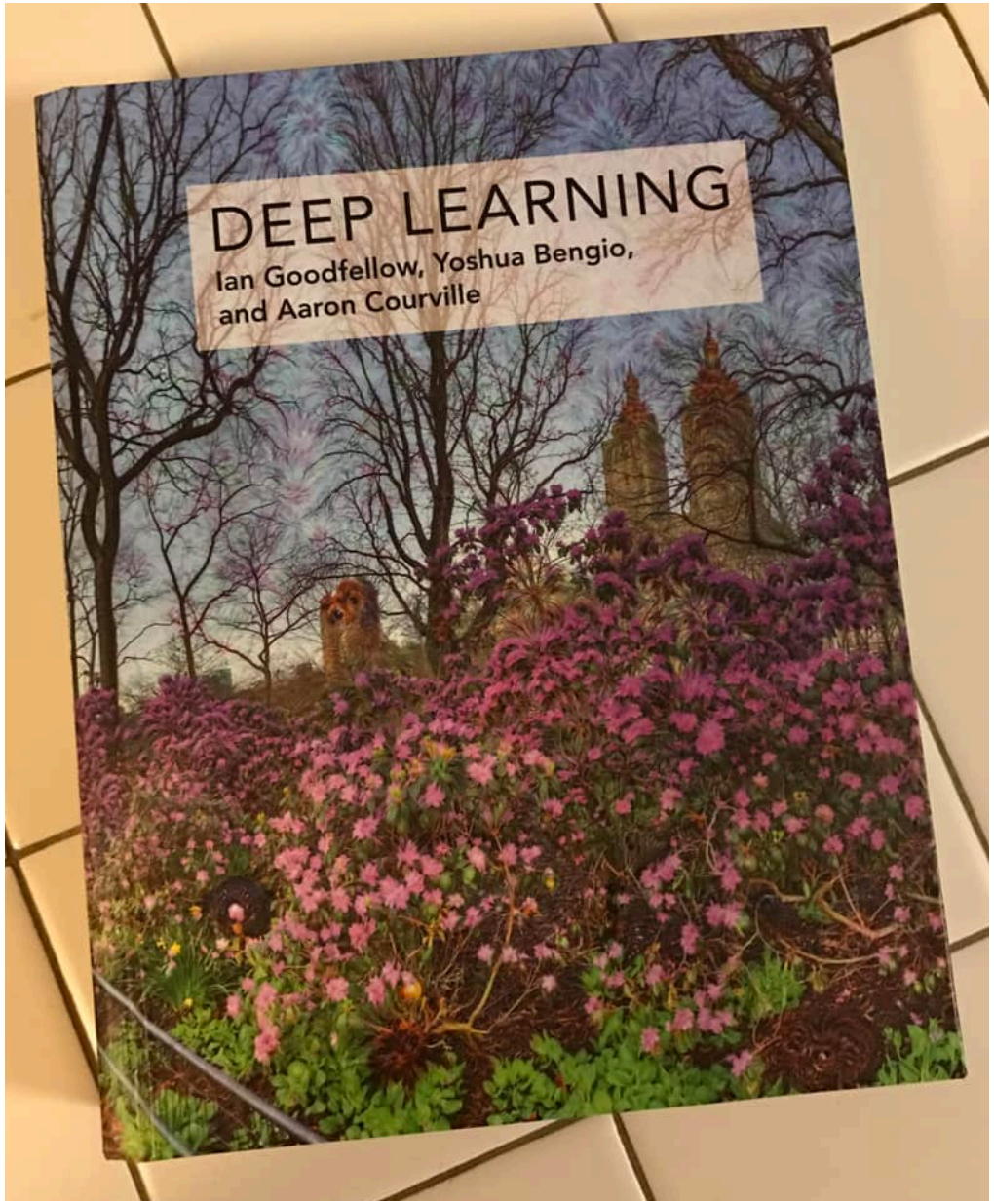
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- Explain what you have tried so far, and what happened
- See StackOverflow guide on asking good questions: https://stackoverflow.com/help/how-to-ask

- Don't expect and answer within 30 minutes of posting
- Monday – Friday, 10am – 6pm EST we'll try to answer within 2 hours
- Other times, we'll try to answer within 12 hours

Optional Textbook

- *Deep Learning* by Goodfellow, Bengio, and Courville
- Free online



Course Content and Grading

- 6 programming assignments (A1 10%, A2-A6 12%)
  - Homework assignments will use Python, PyTorch, and Google Colab
- Midterm Exam (30%)
- Late policy
  - 3 free late days to use on assignments
  - Once free late days are exhausted, 25% penalty per day

Collaboration Policy

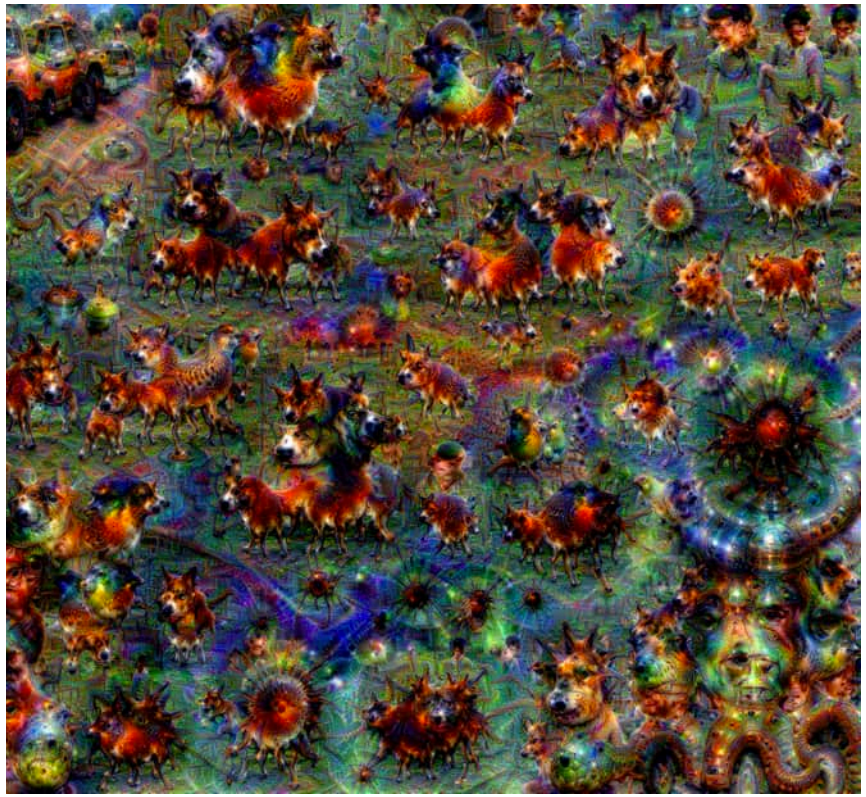
- **Rule 1**: Don't look at solutions or code that are not your own; everything you submit should be your own work
- **Rule 2**: Don't share your solution code with others; however discussing ideas or general strategies is fine and encouraged
- **Rule 3**: Indicate in your submissions anyone you worked with
- Turning in something late / incomplete is better than violating the honor code

Course Philosophy

- Thorough and Detailed.
  - This not “Learn PyTorch in 90 days”, nor “Deep Learning in 10 lines of code”
  - Understand how to write from scratch, debug, and train convolutional and other types of deep neural networks
  - We prefer to write from scratch, rather than rely on existing implementations
- Practical
  - Focus on practical techniques for training and debugging neural networks
  - Will use state-of-the-art software tools like PyTorch and TensorFlow
- State of the art
  - Most material we cover is research published in the last 5 years

Course Philosophy

- Will also cover some fun topics:
  - Image captioning
  - DeepDream, Artistic Style Transfer







Course Structure

- First half: Fundamentals
  - Details of how to implement and train different types of networks
  - Fully-connected networks, convolutional networks, recurrent networks
  - How to train and debug, very detailed
- Second half: Applications and “Researchy” topics
  - Object detection, image segmentation, 3D vision, videos
  - Attention, Transformers
  - Vision and Language
  - Generative models: GANs, VAEs, etc
  - Guest Lectures from subject-matter experts
  - Less detailed: provide overview and references, but skip some details

First homework assignment

- Will be released by tomorrow
- Due Friday 9/11/2020
- Next lecture will be enough to complete it

Next time: Image Classification